V93000 Specifications

V93000 Device Power Supplies

Version 8.6.0

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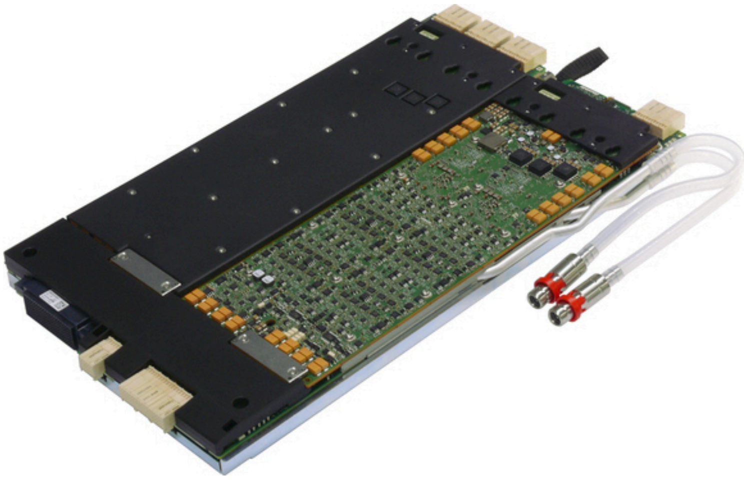
DC Scale AVI64 Universal Pin VI (E8025AVI)

The AVI64 is a general purpose V/I card that features analog and high-voltage digital capabilities and is optimized for providing a true universal analog pin, covering a wide range of test application needs. The card belongs to the DC Scale family and can reside in any channel card slot.

Scope of specifications

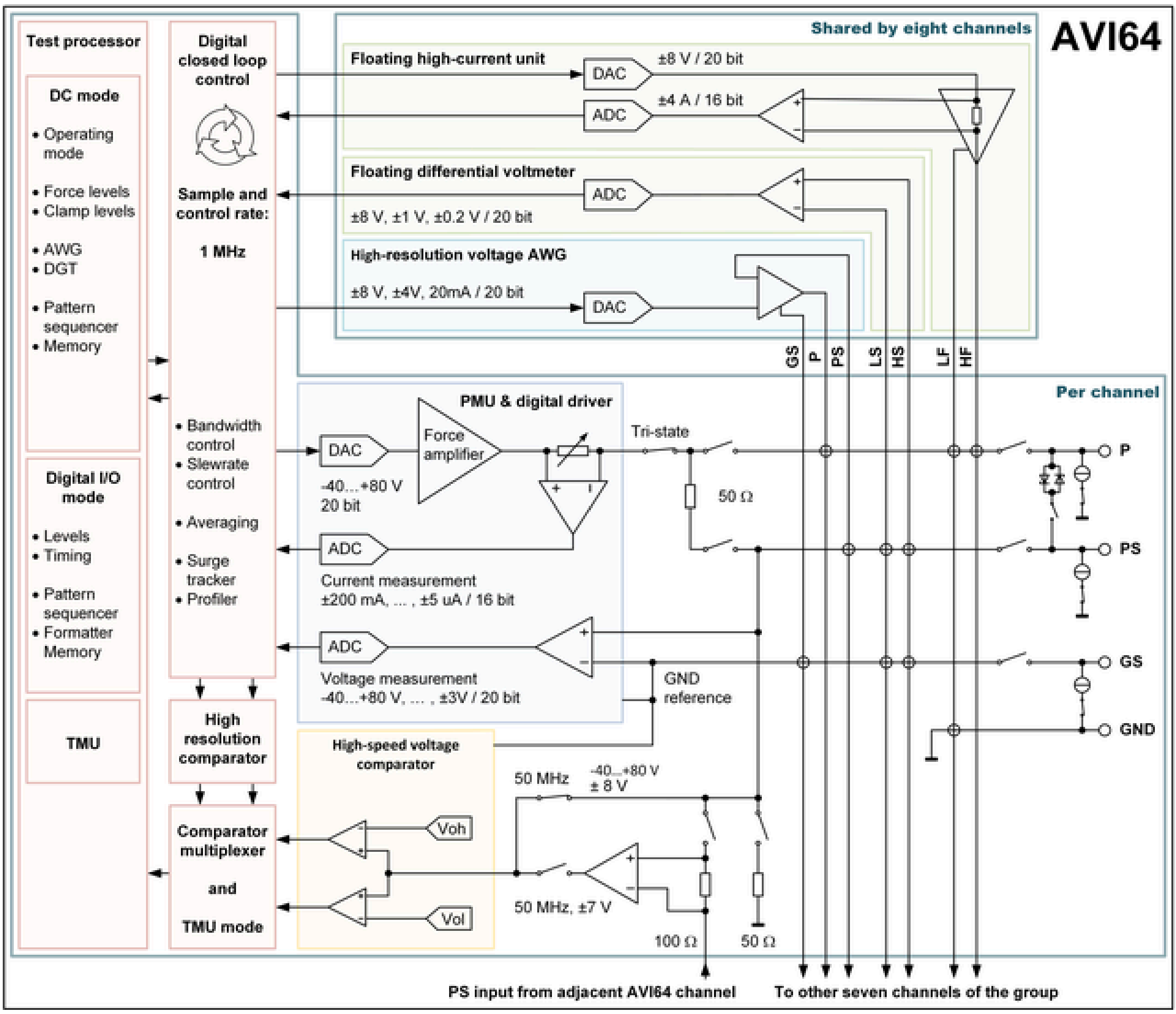
Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.



AVI64 card photo

AVI64 block diagram and configuration

Block diagram



AVI64 block diagram

Configuration		
Number of VI channels per card	64 (8 groups, 8 channels each)	
Maximum source / sink current per card	Voltage ranges	Maximum current per card in range
	-40 V ... +80 V	0.8 A
	±30 V	3.2 A
	±10 V (±3 V)	12.8 A[1]

[1] *Total current for all channels in both ranges.*

Configuration				
	Maximum total current for any given range is fully independent of total current used for other voltage ranges on the same card.			
Maximum number of cards and channels	A -test head	Compact test head	Small test head	Large test head
Maximum number of cards per system	8	16	32	64
Maximum number of VI channels per system	512	1024	2048[2]	4096

Other configuration data

- AVI64 is slot compatible to AVI64, DPS64, DPS128, PVI8, and Pure Clock cards.
- Diagnostics requires the Diagnostics Interconnect Boards E7010E, E7010F, or E7010MUX.

[2] Using more than 1024 channels requires the "DUT Scale Interface".

AVI64 PMU voltage force and measurement specifications

Voltage force[3][4]	Range[5]	Resolution	Accuracy
	±3 V	10 µV	± (25 µV + 0.003 % of relative setting)[6][7]
			± (100 µV + 0.003 % of setting)[6]
			± (500 µV + 0.01 % of setting)
	±10 V	25 µV	± (80 µV + 0.003 % of relative setting)[6][7]
			± (300 µV + 0.003 % of setting)[6]
			± (1.0 mV + 0.01 % of setting)
	±30 V	70 µV	± (350 µV + 0.01 % of relative setting)[6][7]
			± (3.5 mV + 0.01 % of setting)
	-40 V ... +80 V	200 µV	± (700 µV + 0.01 % of relative setting)[6][7]
			± (10 mV + 0.01 % of setting)

Voltage measurement[3]	Range[8]	Resolution	Accuracy[9]
	±3 V	10 µV	± (25 µV + 0.003 % of relative reading)[10][11]
			± (100 µV + 0.003 % of reading)[10]
			± (500 µV + 0.01 % of reading)
	±10 V	25 µV	± (80 µV + 0.003 % of relative reading)[10][11]
			± (300 µV + 0.003 % of reading)[10]
			± (1.0 mV + 0.01 % of reading)
	±30 V	70 µV	± (350 µV + 0.01 % of relative reading)[10][11]
			± (3.5 mV + 0.01 % of reading)

[3] *GND sense can be ±5 V around system ground. GND sense voltage reduces usable voltage range except for ±3 V range.*
[4] *Maximum allowed voltage drop on force line: ±1 V.*
[5] *Voltage ranges can be selected independent of programmed current range.*
[6] *DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days.*
[7] *Specifications are valid within 6 hours. Requires hardware revision 008 or later.*
[8] *Voltage measurement range is controlled by voltage force settings (or voltage clamp, if active).*
[9] *Requires 1000 averaging samples.*
[10] *DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days. Specifications are for averaging over 1 power line cycle.*
[11] *Specifications are valid for measurements taken within 6 hours. Requires hardware revision 008 or later.*

Voltage measurement[3]	**Range**[8]	**Resolution**	**Accuracy**[9]
	-40 V ... +80 V	200 µV	± (700 µV + 0.01 % of relative reading)[10][11] ± (10 mV + 0.01 % of reading)

Current clamp	**Range**	**Resolution**	**Accuracy**[12]
	±200 mA	10 µA	± (500 µA + 0.2 % of setting + 1 µA / V * $\lvert V_{act} \rvert$)
	±50 mA	2 µA	± (50 µA + 0.2 % of setting + 0.2 µA / V * $\lvert V_{act} \rvert$)
	±5 mA	200 nA	± (5 µA + 0.2 % of setting + 20 nA / V * $\lvert V_{act} \rvert$)
	±500 µA	20 nA	± (0.5 µA + 0.2 % of setting + 2 nA / V * $\lvert V_{act} \rvert$)
	±50 µA	2 nA	± (50 nA + 0.2 % of setting + 1 nA / V * $\lvert V_{act} \rvert$)
	±5 µA	200 pA	± (10 nA + 0.2 % of setting + 1 nA / V * $\lvert V_{act} \rvert$)
n channels ganged[13]	±n * 200 mA	n * 10 µA	Sum of single channel errors

[8] *Voltage measurement range is controlled by voltage force settings (or voltage clamp, if active).*
[9] *Requires 1000 averaging samples.*
[12] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the AVI64 channel is in (forcing or clamping).*
[13] *Unlimited ganging of all channels on a single card requires software version 8.1.0 or later. Otherwise, ganging of up to 8 consecutive channels in same channel group is supported by software.*

AVI64 PMU current force and measurement specifications

Current force	Range[14]	Resolution	Accuracy[15]
	±200 mA	10 µA	± (100 µA + 0.05 % of setting + 1 µA / V * $\lvert V_{act}\rvert$)
	±50 mA	2 µA	± (25 µA + 0.05 % of setting + 0.2 µA / V * $\lvert V_{act}\rvert$)
	±5 mA	200 nA	± (2.5 µA + 0.05 % of setting + 20 nA / V * $\lvert V_{act}\rvert$)
	±500 µA	20 nA	± (0.25 µA + 0.05 % of setting + 2 nA / V * $\lvert V_{act}\rvert$)
	±50 µA	2 nA	± (25 nA + 0.05 % of setting + 1 nA / V * $\lvert V_{act}\rvert$)
	±5 µA	200 pA	± (5 nA + 0.05 % of setting + 1 nA / V * $\lvert V_{act}\rvert$)
n channels ganged[16]	±n * 200 mA	n * 10 µA	Sum of single channel errors

Current measurement	Range[17]	Resolution	Accuracy[15][18]
	±200 mA	10 µA	± (100 µA + 0.05 % of reading + 1 µA / V * $\lvert V_{act}\rvert$)
	±50 mA	2 µA	± (25 µA + 0.05 % of reading + 0.2 µA / V * $\lvert V_{act}\rvert$)
	±5 mA	200 nA	± (2.5 µA + 0.05 % of reading + 20 nA / V * $\lvert V_{act}\rvert$)
	±500 µA	20 nA	± (0.25 µA + 0.05 % of reading + 2 nA / V * $\lvert V_{act}\rvert$)
	±50 µA	2 nA	± (25 nA + 0.05 % of reading + 1 nA / V * $\lvert V_{act}\rvert$)
	±5 µA	200 pA	± (5 nA + 0.05 % of reading + 1 nA / V * $\lvert V_{act}\rvert$)
n channels ganged[16]	±n * 200 mA	n * 10 µA	Sum of single channel errors

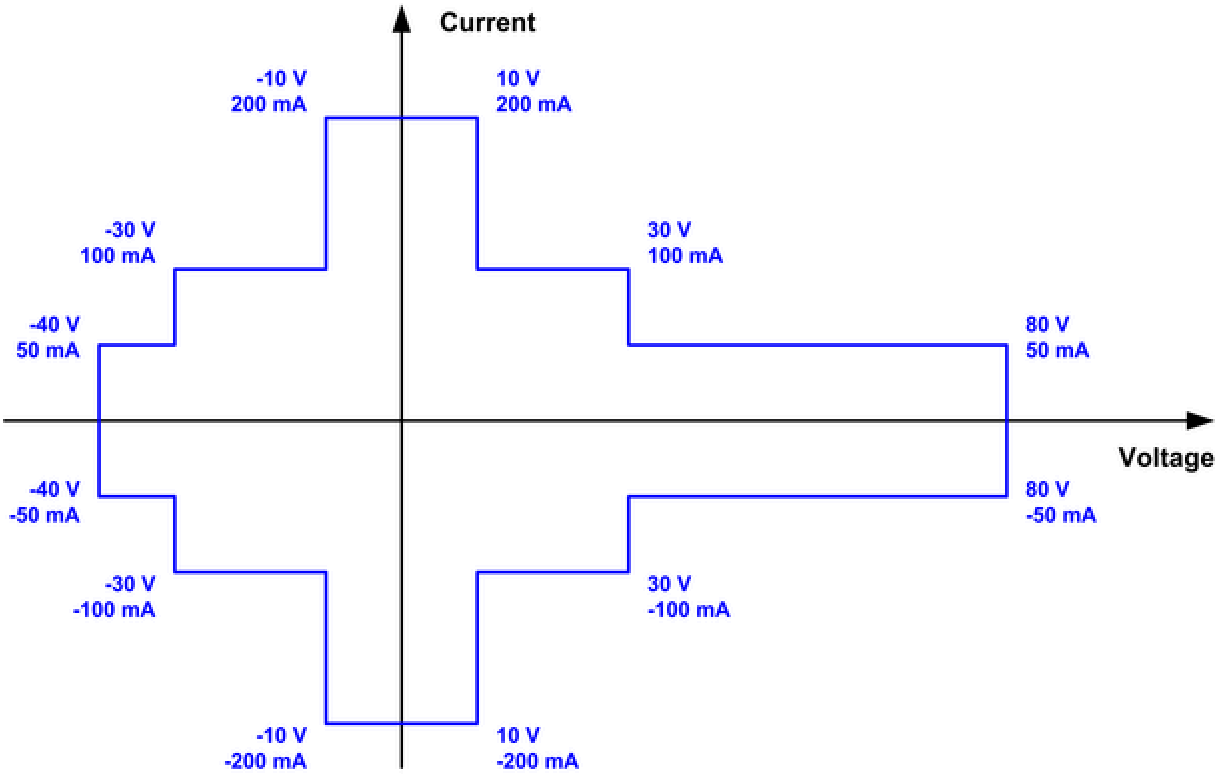
Voltage clamp	Range	Resolution	Accuracy
	±3 V	10 µV	± (3 mV + 0.1 % of setting)

[14] *Current ranges can be selected independent of programmed voltage range.*
[15] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).*
[16] *Unlimited ganging of all channels on a single card requires software version 8.1.0 or later. Otherwise, ganging of up to 8 consecutive channels in same channel group is supported by software.*
[17] *Current measurement range is controlled by current force settings (or current clamp, if active).*
[18] *Requires 1000 averaging samples.*

Voltage clamp	Range	Resolution	Accuracy
	±10 V	25 µV	± (10 mV + 0.1 % of setting)
	±30 V	70 µV	± (30 mV + 0.1 % of setting)
	-40 V ... +80 V	200 µV	± (80 mV + 0.1 % of setting)

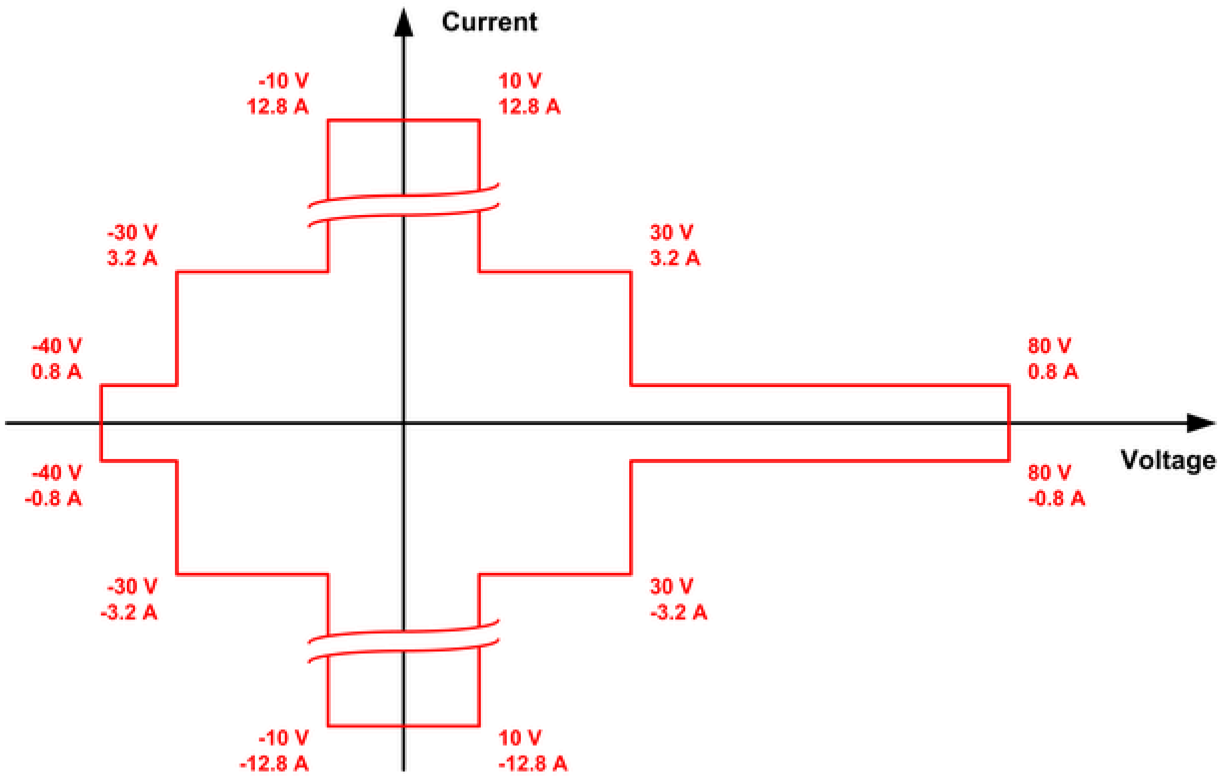
AVI64 PMU voltage and current capability specifications

Voltage and current capabilities per channel



AVI64 PMU voltage and current capability per channel

Maximum current per card and voltage range



AVI64 power budget per card and voltage range

AVI64 digitizer specifications

Per-pin voltage digitizer[19]

Voltage ranges	± 3 V	± 10 V	± 30 V	-40 V ... +80 V
Resolution	20 bit (10 µV)	20 bit (25 µV)	20 bit (70 µV)	20 bit (200 µV)
DC accuracy[20]	±(0.5 mV + 100 ppm of reading)	±(1 mV + 100 ppm of reading)	±(3.5 mV + 100 ppm of reading)	±(10 mV + 100 ppm of reading)
INL (typical)	± 24 µV (± 4 ppm FSR)	± 40 µV (± 2 ppm FSR)	± 500 µV (± 8 ppm FSR)	± 1 mV (± 8 ppm FSR)
DNL (typical)	± 1 ppm FSR			
Ground sense range[21]	± 5 V[22]			
Input impedance (typical)	> 1 GΩ			
Input capacitance (typical)	165 pF			
Sampling rate	0 ... 1 Msps (million samples per second)			
Analog bandwidth (typical, -3 dB)	250 kHz			
Filters	Programmable digital filter 100 Hz ... 250 kHz			
Waveform capture memory (per pin)	Software configurable, not exceeding total available memory size per channel (107 MByte)[23], 32 Bytes required per sample.			

		± 3 V range	± 10 V range
Noise (typical)	22 Hz ... 22 kHz	30 μV_{RMS}	50 μV_{RMS}
	22 Hz ... 100 kHz	35 μV_{RMS}	150 μV_{RMS}
	22 Hz ... 250 kHz	45 μV_{RMS}	180 μV_{RMS}
SNR (typical)	1 kHz[24]	98 dB @ 5 V_{PP}	94 dB @ 8 V_{PP}
	10 kHz[25]	94 dB @ 5 V_{PP}	89 dB @ 8 V_{PP}
	20 kHz[25]	93 dB @ 5 V_{PP}	88 dB @ 8 V_{PP}

[19] Voltage and current digitizer are available for every channel and can be used in parallel. Software selectable range settings are the same as for PMU modes.
[20] 1 kHz digital filter.
[21] Independent GND sense per channel.
[22] GND sense voltage reduces usable voltage range except for ± 3 V range.
[23] Memory pooling is supported for every eight channels in the same channel group.
[24] 22 kHz digital filter, bandwidth 22 Hz ... 22 kHz, 1 Msps, initial discard 120 samples.
[25] 100 kHz digital filter, bandwidth 22 Hz ... 100 kHz, 1 Msps, initial discard 30 samples.

Per-pin voltage digitizer[19]			
THD (typical)	1 kHz[26]	-105 dB @ 5 V_{PP}	-102 dB @ 8 V_{PP}
	10 kHz[27]	-98 dB @ 5 V_{PP}	-92 dB @ 8 V_{PP}
	20 kHz[27]	-97 dB @ 5 V_{PP}	-92 dB @ 8 V_{PP}

Current digitizer[19]			
Current ranges[28]	±5 µA, ±50 µA, ±500 µA, ±5 mA, ±50 mA, ±200 mA		
Resolution	16 bit		
Sampling rate	0 ... 1 Msps (million samples per second)		
Analog bandwidth (-3 dB)	250 kHz (typical)		
Programmable digital filter	100 Hz to 250 kHz		
Waveform capture memory (per pin)	Software configurable, not exceeding total available memory size per channel (107 MByte)[23], 32 Bytes required per sample.		
DC Accuracy	**Range[29]**	**Resolution**	**Accuracy[20][30]**
	±200 mA	10 µA	± (100 µA + 0.05 % of reading + 1 µA / V * |V_{act}|)
	±50 mA	2 µA	± (25 µA + 0.05 % of reading + 0.2 µA / V * |V_{act}|)
	±5 mA	200 nA	± (2.5 µA + 0.05 % of reading + 20 nA / V * |V_{act}|)
	±500 µA	20 nA	± (0.25 µA + 0.05 % of reading + 2 nA / V * |V_{act}|)
	±50 µA	2 nA	± (25 nA + 0.05 % of reading + 1 nA / V * |V_{act}|)
	±5 µA	200 pA	± (5 nA + 0.05 % of reading + 1 nA / V * |V_{act}|)
n channels ganged[31]	±n * 200 mA	n * 10 µA	Sum of single channel errors

[19] Voltage and current digitizer are available for every channel and can be used in parallel. Software selectable range settings are the same as for PMU modes.
[26] 22 kHz digital filter, H2 ... H5, 1 Msps, initial discard 120 samples.
[27] 100 kHz digital filter, H2 ... H5, 1 Msps, initial discard 30 samples.
[28] Current measure range is used in digitizer mode.
[29] Current measurement range is controlled by current force settings (or current clamp, if active).
[30] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).
[31] Unlimited ganging of all channels on a single card requires software version 8.1.0 or later. Otherwise, ganging of up to 8 consecutive channels in same channel group is supported by software.

AVI64 AWG specifications

Per-pin voltage AWG			
Voltage ranges[32]		±3 V, ±10 V, ±30 V, -40 V … +80 V	
Resolution		20 bit	
DC Accuracy		See Voltage force accuracy	
Maximum current		See Voltage / Current capabilities	
Ground sense range[33]		± 5 V[34]	
Sampling rate		> 0 … 1 Msps (million samples per second)	
Analog bandwidth (typical)		20 kHz	
Waveform memory (per pin)[35]		1 Msamples	
		± 3 V range	± 10 V range
Stability (typical)		± 18 µV (± 3 ppm FSR)	± 40 µV (± 2 ppm FSR)
INL (typical)		± 30 µV (± 5 ppm FSR)	± 100 µV (± 5 ppm FSR)
DNL (typical)		± 0.5 ppm FSR	± 0.5 ppm FSR
Noise (typical)	22 Hz … 22 kHz[36]	420 μV_{RMS}	450 μV_{RMS}
	22 Hz … 100 kHz[36]	680 μV_{RMS}	700 μV_{RMS}
SNR (typical)	1 kHz[37]	65 dB @ 5 V_{PP}	70 dB @ 8 V_{PP}
THD (typical)	1 kHz[38]	-60 dBc @ 5 V_{PP}	-65 dBc @ 8 V_{PP}

Per-pin current AWG[39]	
Current ranges[40]	±5 µA, ±50 µA, ±500 µA, ±5 mA, ±50 mA, ±200 mA
Resolution	16 bit
Accuracy	See Current force accuracy
Sampling rate	0 … 1 Msps (million samples per second)

[32] *Voltage force range is used in AWG mode.*
[33] *Independent GND sense per channel.*
[34] *GND sense voltage reduces usable voltage range except for ± 3 V range.*
[35] *Waveform memory shared between all AWG modes and DC test points.*
[36] *200 mA current range selected.*
[37] *Bandwidth 22 Hz … 22 kHz, 1000 ksps.*
[38] *H2 … H5, 1000 ksps*
[39] *Voltage and current AWG are available for every channel. Software selectable range settings.*
[40] *Current force range is used in AWG mode.*

Per-pin current AWG[39]	
Analog bandwidth (typical)	20 kHz (50 µA ... 200 mA range)
Waveform memory (per pin)[35]	1 Msamples

High-speed voltage AWG[41]	
Voltage ranges	± 8 V, ± 30 V, -40 V ... +80 V
Resolution	14 bit (1 mV @ ± 8 V, 10 mV @ ± 30 V, 10 mV @ -40 V ... + 80 V)
Accuracy	± (50 mV + 0.1 % of setting)[42]
Sampling rate	0 ... 10 Msps (million samples per second)
Analog bandwidth (typical)	1 MHz[42]
Maximum current	200 mA @ ± 8 V, 100 mA @ ± 30 V, 50 mA @ -40 V ... + 80 V
Source impedance (typical)	10 Ω / 50 Ω[42][43]
Waveform memory (per pin)[35]	1 Msamples

High-resolution voltage AWG[44]		
	± 4 V range	± 8 V range
Resolution	20 bit (10 µV)	20 bit (20 µV)
Accuracy	± (50 µV + 30 ppm of relative setting)[45][46]	± (80 µV + 30 ppm of relative setting)[45][46]
	± (100 µV + 30 ppm of setting)[45]	± (200 µV + 30 ppm of setting)[45]
	± (150 µV + 100 ppm of setting)	± (300 µV + 100 ppm of setting)
Stability (typical)	± 10 µV (± 1.25 ppm FSR)	± 20 µV (± 1.25 ppm FSR)
Maximum current	± 20 mA	± 20 mA
Ground sense range[33]	± 0.5 V	± 0.5 V
INL (typical)	± 32 µV (± 4 ppm FSR)	± 64 µV (± 4 ppm FSR)
DNL (typical)	± 1 ppm FSR	± 1 ppm FSR

[39] Voltage and current AWG are available for every channel. Software selectable range settings.
[41] High speed AWG is available for every channel. Software selectable range settings.
[42] Valid in 200 mA range.
[43] 10 Ω when connected over force line (P#), 50 Ω when connected over sense line (PS#).
[44] High Resolution AWG is available once for every group of eight channels. It can be enabled by software for any channel of the group and replaces the PMU of the channel when enabled.
[45] DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days.
[46] Specifications are valid within 6 hours. Requires hardware revision 008 or later.

High-resolution voltage AWG[44]			
Noise (typical)	22 Hz … 22 kHz	10 μV_{RMS}	10 μV_{RMS}
	22 Hz … 100 kHz	20 μV_{RMS}	20 μV_{RMS}
SNR (typical)	50 / 60 Hz[47]	114 dB @ 7 V_{PP}	117 dB @ 14 V_{PP}
	1 kHz[48]	106 dB @ 7 V_{PP}	112 dB @ 14 V_{PP}
	1 kHz[48] (EIAJ A-weighted)	110 dB @ 7 V_{PP}	116 dB @ 14 V_{PP}
	10 kHz[49]	98 dB @ 7 V_{PP}	103 dB @ 14 V_{PP}
	20 kHz[49]	98 dB @ 7 V_{PP}	98 dB @ 14 V_{PP}
THD (typical)	50 / 60 Hz[50]	-110 dBc @ 7 V_{PP}	-113 dBc @ 14 V_{PP}
	1 kHz[51]	-96 dBc @ 7 V_{PP}	-94 dBc @ 14 V_{PP}
	10 kHz[52]	-73 dBc @ 7 V_{PP}	-70 dBc @ 14 V_{PP}
	20 kHz[52]	-73 dBc @ 7 V_{PP}	-64 dBc @ 14 V_{PP}
Sampling rate		0 … 1 Msps (million samples per second)	
Analog bandwidth (typical)		125 kHz	
Selectable hardware filters		1.5 kHz / 25 kHz / bypass	
Waveform memory (per pin)[35]		1 Msamples	

[44] *High Resolution AWG is available once for every group of eight channels. It can be enabled by software for any channel of the group and replaces the PMU of the channel when enabled.*
[47] *1.5 kHz hardware filter selected, bandwidth 22 Hz … 2 kHz, 10 ksps.*
[48] *1.5 kHz hardware filter selected, bandwidth 22 Hz … 22 kHz, 200 ksps.*
[49] *25 kHz hardware filter selected, bandwidth 22 Hz … 100 kHz, 1000 ksps.*
[50] *1.5 kHz hardware filter selected, H2 … H5, 10 ksps.*
[51] *1.5 kHz hardware filter selected, H2 … H5, 200 ksps.*
[52] *25 kHz hardware filter selected, H2 … H5, 1000 ksps.*

AVI64 floating differential voltmeter specifications

There are eight floating differential voltmeters for every card (one per group of eight channels). Floating differential voltmeters can be used for (software selectable) differential measurements on any channel of that group or between two (software selectable) channels of such a group. The floating differential voltmeters can also be used as floating differential digitizer.

When using a floating high-current unit, the floating differential voltmeter of that group will be used for measuring voltages at the floating channel and cannot be used for other measurements on the same group of eight channels.

Floating differential voltmeter / digitizer	
Voltage ranges	±200 mV, ±1 V, ±8 V
Common mode range[53]	-40 V ... +80 V
Input current (maximum)	± (1 nA + 1 nA / V * $\lvert V_{act} \rvert$)[54]
Input capacitance (typical)	530 pF (differential), 750 pF (single ended to GND)

Voltage measurement	**Range**	**Resolution**	**Accuracy[55]**
	±200 mV	0.4 µV	± (10 µV + 0.003 % of relative reading)[56][57]
			± (60 µV + 0.003 % of reading)[56]
			± (200 µV + 0.05 % of reading)
	±1 V	2 µV	± (15 µV + 0.003 % of relative reading)[56][57]
			± (100 µV + 0.003 % of reading)[56]
			± (300 µV + 0.01 % of reading)
	±8 V	20 µV	± (50 µV + 0.003 % of relative reading)[56][57]
			± (300 µV + 0.003 % of reading)[56]
			± (1.0 mV + 0.01 % of reading)

Floating differential digitizer			
	± 200 mV range	± 1 V range	± 8 V range
Resolution	20 bit (0.4 µV)	20 bit (2 µV)	20 bit (20 µV)

[53] *The voltage at either input of the differential voltmeter must not exceed common mode range.*
[54] *V_{act} is the actual voltage at either input of the differential voltmeter.*
[55] *Requires 1000 averaging samples.*
[56] *DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days. Specifications are for averaging over 1 power line cycle.*
[57] *Specifications are valid for measurements taken within 6 hours. Requires hardware revision 008 or later.*

Floating differential digitizer			
DC Accuracy[58]	±(0.2 mV + 500 ppm of reading)	±(0.3 mV + 100 ppm of reading)	±(1 mV + 100 ppm of reading)
INL (typical)	± 3 µV (± 7 ppm FSR)	± 6 µV (± 3 ppm FSR)	± 40 µV (± 3 ppm FSR)
DNL (typical)	± 1 ppm FSR		
Sampling rate	0 … 1 Msps (million samples per second)		
Analog bandwidth (typical, -3 dB)	250 kHz		
Filters	Programmable digital filter 100 Hz ... 250 kHz		
Waveform capture memory (per pin)	Software configurable, not exceeding total available memory size per channel (107 MByte)[59], 32 Bytes required per sample.		

[58] *1 kHz digital filter.*
[59] *Memory pooling is supported for every eight channels in the same channel group.*

AVI64 floating high-current unit (HCU) specifications

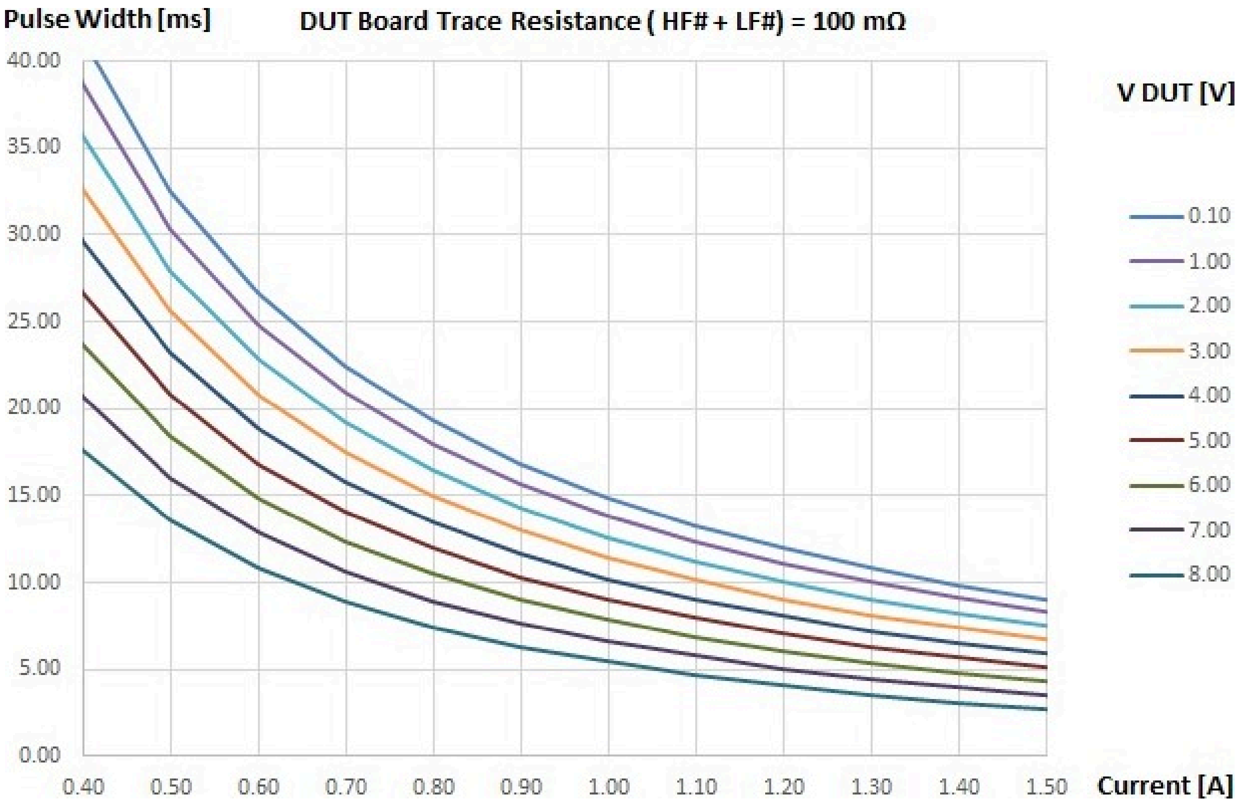
There are eight floating high-current units for every card (one per group of eight channels). Floating high-current capabilities can be added to any pin using software selection.

Floating high-current unit (pulsed operation)	
Voltage range	±8 V
Voltage measure resolution	20 µV
Voltage measure accuracy	± (1 mV + 0.01 % of reading)[60]
Voltage force resolution	300 µV
Voltage force accuracy	± (3.5 mV + 0.03 % of setting)
Voltage clamp accuracy	± (10 mV + 0.1 % of setting)
Floating range	-30 V ... +80 V[61]
Current range	±4 A (pulsed operation)
Current measure resolution	140 µA
Current measure accuracy	± (2 mA + 0.1 % of reading)[60][62]
Current force resolution	200 µA
Current force accuracy	± (2 mA + 0.1 % of setting)[62]
Current clamp accuracy	± (4 mA + 0.5 % of setting)
Minimum pulse width	100 µs
Maximum pulse width	See diagrams below

Pulse width vs. load current - 100 mΩ DUT board resistance

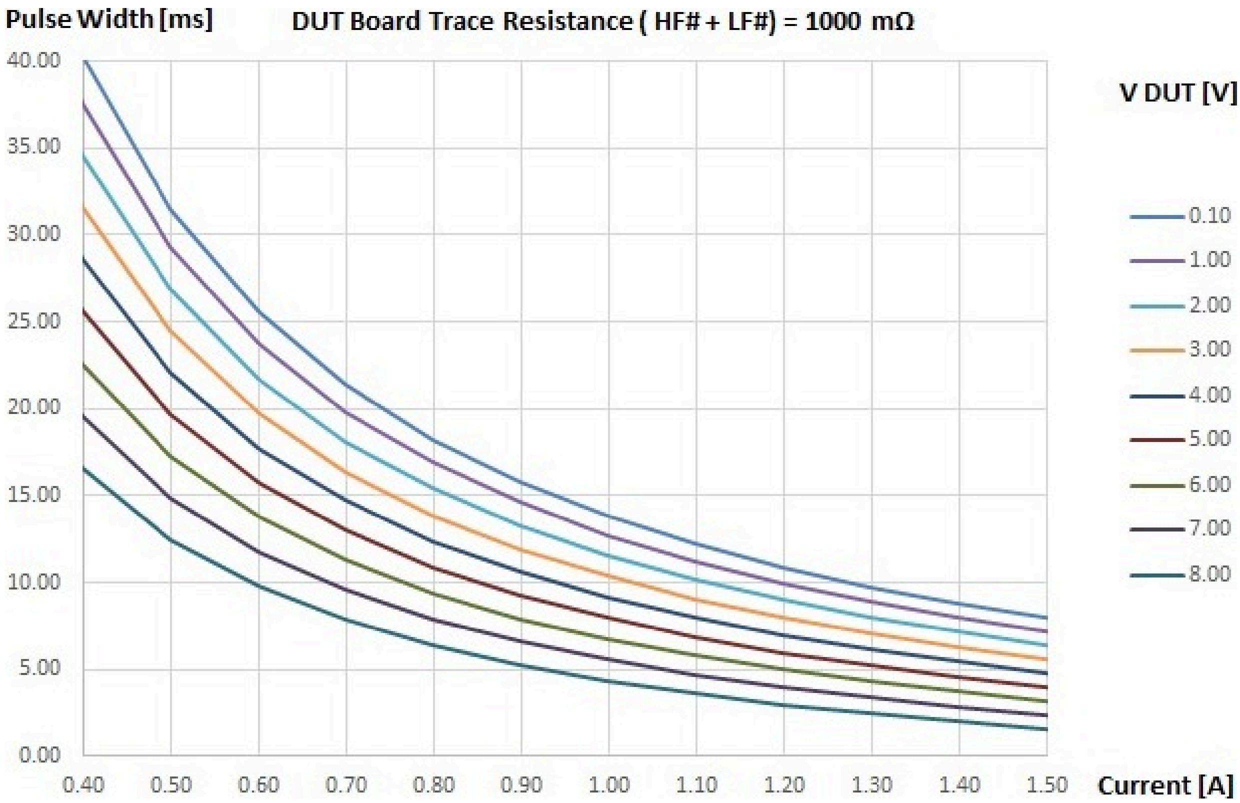
The following two diagrams show the relationship between the current, the voltage at the DUT, and the maximum pulse length for a total DUT board trace resistance of 100 mΩ. Note: The total DUT board trace resistance includes the resistance of the HF# traces, the LF# traces, and the contact resistance at the DUT.

[60] *20 kHz low pass filter. Requires 100 averaging samples.*
[61] *The voltage at either of the force and sense pins must not exceed floating range.*
[62] *Valid if calibrated with software version 7.4.1 or later.*



Pulse width vs. load current - 1000 mΩ DUT board resistance

The following two diagrams show the relationship between the current, the voltage at the DUT, and the maximum pulse length for a total DUT board trace resistance of 1000 mΩ. Note: The total DUT board trace resistance includes the resistance of the HF# traces, the LF# traces, and the contact resistance at the DUT.


Pulse Width [ms]
DUT Board Trace Resistance (HF# + LF#) = 1000 mΩ
V DUT [V]
0.10
1.00
2.00
3.00
4.00
5.00
6.00
7.00
8.00
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
0.40
0.50
0.60
0.70
0.80
0.90
1.00
1.10
1.20
1.30
1.40
1.50
Current [A]


Pulse Width [ms]
DUT Board Trace Resistance (HF# + LF#) = 1000 mΩ
V DUT [V]
0.10
1.00
2.00
3.00
4.00
5.00
6.00
7.00
8.00
8.00
7.00
6.00
5.00
4.00
3.00
2.00
1.00
0.00
1.50
2.00
2.50
3.00
3.50
4.00
Current [A]

AVI64 digital driver, comparator, and TMU specifications

Digital I/O operation mode

The Digital I/O operation mode can be selected on a per channel granularity. Changing the mode between the "Digital I/O operating mode" and the "PMU operating mode" (and vice versa) within a test program flow is supported independently for every channel.

Digital I/O operation mode	
Vector period range	2.5 ns ... 31250 ns
Vector period accuracy	±15 ppm of period setting
Vector period resolution	0.001 fs
Edge placement range[63]	-1 ... 32 vector periods
Edge placement resolution	1 ps
Total available memory per channel	107 MByte

Digital driver[64]			
		$-8\text{ V} \le V_{IL}, V_{IH}, V_T \le +8\text{ V}$	$-40\text{ V} \le V_{IL}, V_{IH}, V_T \le +80\text{ V}$
Driver states		V_{IL}, V_{IH}, V_T[65], High-Z	V_{IL}, V_{IH}, V_T[65], High-Z
Level accuracy		± (30 mV + 0.1 % of setting)	± (50 mV + 0.1 % of setting)
Overshoot / undershoot		10 % of programmed level	10 % of programmed level
Maximum drive current		± 50 mA	± 50 mA
Driver slew rate (typical)		50 V / µs @ 3 V swing	80 V / µs @ 12 V swing
Minimum pulse width		100 ns at 3 V swing	400 ns at 12 V swing
Maximum driver swing		16 V	32 V
Maximum data rate (20% / 80%); tr, tf < (Tcycle / 2):[66]			
	3V swing	10 Mbps	5 Mbps
	5V swing	5 Mbps	5 Mbps
	12V swing	2.5 Mbps	2.5 Mbps
	24V swing	n/a	1 Mbps

[63] Maximum edge placement range is -6250 ns to +19000 ns. For vector periods > 752 ns the edge placement range is -1 to 12 vector periods. For vector periods < 30 ns the edge placement range is +60 ns to 32 vector periods.

[64] Specifications are given for having "sense line" connection only.

[65] V_T is the third level which can be used instead of "High-Z". The third level is software programmable within the maximum driver level range independent of V_{IL} and V_{IH}.

[66] Programmable up to 20 Mbit/s.

Digital driver[64]		
Delay time active # high-Z	10 µs (maximum)	10 µs (maximum)
Edge placement accuracy (EPA)	± 10 ns[67]	± 10 ns[67]
Digital compare (high-speed mode)		
Programmable thresholds[68]	V_{OL}, V_{OH}	
Input voltage and threshold ranges (single ended)[69]	-8 V ≤ V_{OL}, V_{OH} ≤ +8 V:	± (30 mV + 0.1 % of setting)
	-40 V ≤ V_{OL}, V_{OH} ≤ +80 V:	± (100 mV + 0.1 % of setting)
Threshold hysteresis (single ended)	25 mV (typical)[70]	
Input voltage range (differential)	-8 V ≤ V_{DUT} ≤ +8 V[71]	
Threshold range (differential)	-7 V ≤ V_{DIFF} ≤ + 7 V:	± (30 mV + 5 % of setting)
Threshold hysteresis (differential)	25 mV (typical)[70]	
Input termination[72]	High impedance,[73] 50 Ω single ended, 100 Ω differential[74]	
Signal path bandwidth	50 MHz (typical)[75]	
Maximum data rate	100 Mbps (typical)[75]	
Intrinsic transition time (20% / 80%)	2 ns (typical)[76]	
Compare timing accuracy (EPA)	± 5 ns[67]	
Compare timing resolution	1 ps	
Digital compare (high-resolution mode)[77]		
Programmable thresholds[78]	V_{OL}, V_{OH}, I_{CL}, I_{CH} (with programmable hysteresis)	

[64] *Specifications are given for having "sense line" connection only.*
[67] *Calibrated at 3 V swing, EPA verified at 1.5 V and 4.5 V swing. Specification is valid across full voltage range (-40 V ... +80 V).*
[68] *Dual level comparator (Low, High, Intermediate, X).*
[69] *Software selectable.*
[70] *Software selectable on / off*
[71] *Protected up to -40 V ... +80 V.*
[72] *Software selectable, single ended termination to GND.*
[73] *1 GΩ (typical)*
[74] *20 kΩ (typical) input impedance to ground*
[75] *3 V swing (0 V ... 3 V)*
[76] *3 V swing (0 V ... 3 V, 50 Ω termination)*
[77] *Software selectable for voltage / current real time P/F compare, digital capture, and TMU mode.*
[78] *Dual level comparator (Low, High, Intermediate, X)*

Digital compare (high-resolution mode)[77]		
Threshold ranges (voltage)[79]	± 3 V, ± 10 V, ± 30 V, -40 V ... +80 V	
Threshold accuracy (voltage)	± 3 V	±(5 mV + 0.03 % of setting)
	± 10 V	±(10 mV + 0.03 % of setting)
	± 30 V	±(30 mV + 0.03 % of setting)
	-40 V ... +80 V	±(80 mV + 0.03 % of setting)
Threshold hysteresis (voltage)	Programmable within full range	
Threshold ranges (current)[80]	± 5 µA, ± 50 µA, ± 500 µA, ± 5 mA, ± 50 mA, ± 200 mA	
Threshold accuracy (current)[81]	± 5 µA	±(20 nA + 0.05% of setting + 1 nA / V * $\|V_{act}\|$)
	± 50 µA	±(150 nA + 0.05% of setting + 1 nA / V * $\|V_{act}\|$)
	± 500 µA	±(1.5 µA + 0.05% of setting + 2 nA / V * $\|V_{act}\|$)
	± 5 mA	±(15 µA + 0.05% of setting + 20 nA / V * $\|V_{act}\|$)
	± 50 mA	±(150 µA + 0.05% of setting + 200 nA / V * $\|V_{act}\|$)
	± 200 mA	±(600 µA + 0.05% of setting + 1 µA / V * $\|V_{act}\|$)
Threshold hysteresis (current)	Programmable within full range	
Compare timing accuracy (EPA)	± 500 ns[82]	
Compare timing resolution	5 ns (interpolated)[83]	
Timing skew (V_{OL} vs. V_{OH}, I_{CL} vs. I_{CH})	± 5 ns (maximum)	
Signal path bandwidth[84]	250 kHz (typical)	

Timing Measurement Unit (TMU)

Timing measurements are done using the digital comparator functionality available per channel (see above).

The TMU allows for time delay / skew measurements (V_{OH} / I_{CH} # V_{OH} / I_{CH}, V_{OL} / I_{CL} # V_{OL} / I_{CL}) also across different instrument types (i.e. PinScale1600), frequency / pulse width measurements, and rise / fall time measurements (V_{OL} / I_{CL} # V_{OH} / I_{CH}, V_{OH} / I_{CH} # V_{OL} / I_{CL}).

[77] *Software selectable for voltage / current real time P/F compare, digital capture, and TMU mode.*
[79] *Voltage measure range is used in compare mode.*
[80] *When used in "Digital I/O" operating mode, threshold range is fixed to ±200 mA.*
[81] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).*
[82] *Additional fixed channel delay in high resolution mode of 3 µs.*
[83] *Sampling rate = 1 µs*
[84] *In "Digital I/O" operating mode*

The TMU can be used with all compare modes above. The following table shows available compare modes for channels in "Digital I/O operating mode" or in "PMU operating mode":

Compare modes	**Digital I/O operating mode**	**PMU operating mode**
High speed (single ended)	supported	supported
High speed (differential)	supported	n/a
High resolution (voltage)	supported	supported
High resolution (differential voltmeter)	n/a	supported
High resolution (current)	supported	supported
High resolution (high current unit)	n/a	supported

Timing Measurement Unit (TMU)	
DC / AC Performance	See "Digital compare" above.
Accuracy for rise / fall time measurements (single shot)[85]	± 5 ns
Accuracy for skew / propagation delay measurements (single shot)	± (2 * EPA)
Accuracy for single timestamp to ARM (time zero)	± 35 ps
Maximum sampling rate	50 Msps

[85] *Available in PMU operating mode only.*

AVI64 other specifications

Interface characteristics	Force	Sense	GND sense
Compliance range if disconnected	-44 V ... +84 V	-44 V ... +84 V	-44 V ... +84 V
Input leakage current (typical)[86]	± (5 nA + 1 nA / V * | V_{act}|)[87]	± (1 nA + 1 nA / V * | V_{act}|)[87]	±200 nA
Input capacitance (typical)[88]	1600 pF	110 pF[89]	150 pF

Surge tracker

Voltage or current surges are detected against programmable minimum and maximum values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two alarms/flags indicate if thresholds have been exceeded. These alarms/flags are latched (stored) and can be retrieved any time in the testflow.

For accuracy specifications of the voltage surge tracker see "Digital compare (high-speed mode)". Voltage threshold accuracies are specified in "Input voltage and threshold ranges (single ended)".

For accuracy specifications of the current surge tracker see "Digital compare (high-resolution mode)". Current threshold accuracies are depending on the current measurement range selected and are specified in "Threshold accuracy (current)".

[86] *Measured in high-impedance mode.*
[87] *V_{act} is the actual voltage of the pin, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).*
[88] *Force pin: The input capacitance is specified for the disconnected mode. Sense pin: The specified input capacitance is for both the connected mode and disconnected mode.*
[89] *In Digital I/O operation mode, 165 pF in PMU operation mode.*

AVI64 pogo pin assignment

The E8025AVP pogo cable assembly connects the 64 channels of the DC Scale AVI64 card to the DUT board.

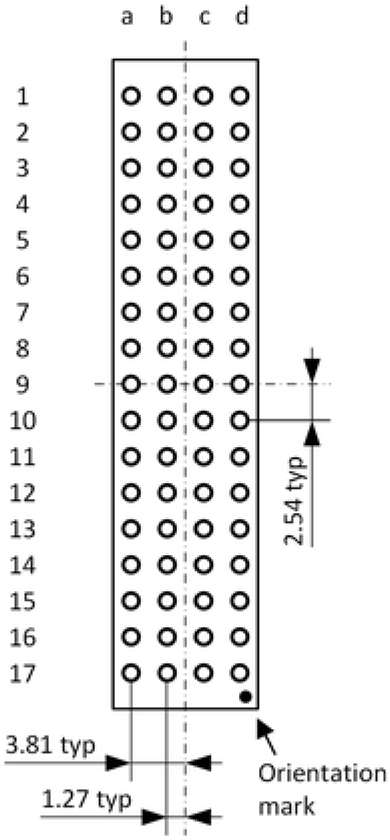
Pogo pin assignment

Block A

	a	b	c	d
1	GS2	P1	GND	PS1
2	PS2	GND	P2	GS1
3	GS4	P3	GND	PS3
4	PS4	GND	P4	GS3
5	GS6	P5	GND	PS5
6	PS6	GND	P6	GS5
7	GS8	P7	GND	PS7
8	PS8	GND	P8	GS7
9	GS10	P9	GND	PS9
10	PS10	GND	P10	GS9
11	GS12	P11	GND	PS11
12	PS12	GND	P12	GS11
13	GS14	P13	GND	PS13
14	PS14	GND	P14	GS13
15	GS16	P15	GND	PS15
16	PS16	GND	P16	GS15
17	DNC	DNC	SAFETY_N	SAFETY_P

Orientation mark •

Layout and dimension



See also *Pogo pad and via design*.

Block B, C, D

	a	b	c	d
1	GS2	P1	GND	PS1
2	PS2	GND	P2	GS1
3	GS4	P3	GND	PS3
4	PS4	GND	P4	GS3
5	GS6	P5	GND	PS5
6	PS6	GND	P6	GS5
7	GS8	P7	GND	PS7
8	PS8	GND	P8	GS7
9	GS10	P9	GND	PS9
10	PS10	GND	P10	GS9
11	GS12	P11	GND	PS11
12	PS12	GND	P12	GS11
13	GS14	P13	GND	PS13
14	PS14	GND	P14	GS13
15	GS16	P15	GND	PS15
16	PS16	GND	P16	GS15
17	DNC	DNC	DNC	DNC

Orientation mark •

Pogo pin description

Pin	Description	DUT board connection
P#	High force line of channel #	
PS#	High sense and digital I/O line of channel #	
GND	Ground line of all channels	Connect to DUT board ground.
GS#	Ground sense line of channel #	
SAFETY_P	Control line of the safety circuit	Connect to an adequate +5.0 V supply via a safety interlock on the DUT board or the DUT interface.
SAFETY_N	Return line of the safety circuit	Connect to the return pin of the +5.0 V supply at SAFETY_P and to the DUT board ground.
DNC	Do not connect pin	These pins are internally used. Do not connect these pins on the DUT board.

Notes on the pogo pin descriptions:

- Pin: # = Channel 1 … 16
- The DNC pins are internally connected and reserved. To ensure that adjacent high voltage pins or traces have no effect on the DNC pins, adequate measures like sufficient insulation space have to be respected.

⚠ CAUTION GND is wired to several pogo pins. All GND pogo pins must be connected in parallel on the DUT board to ensure that the specifications for the card and the maximum ratings for the pogo pins are met.

DC Scale DPS128 Device Power Supply (E8023CS/E8023CSH) - high current specifications

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Configuration

	E8023CSH	**E8023CS**
Maximum number of channels per board	64	128
Maximum source/sink current per channel	1.0 A @ 2.5 V	1.0 A @ 2.5 V
Maximum source/sink current per channel	0.5 A @ 7 V	0.5 A @ 7 V
Maximum source/sink current per board	64 A /64 A@ 2.5 V	66 A /58 A@ 2.5 V
Voltage range	-2.5 V to +7 V	-2.5 V to +7 V

The DPS128/DPS64 card features two power domains with a dedicated power budget. Each domain is able to deliver half of the overall power budget. The power domain to channel mapping must be observed for DPS128 cards when assigning the available channels to the DUT pins. For more information refer to TDC topic DPS128/DPS64 power budget (147811).

	A#Class test head[90]	**Compact test head[90]**	**Small test head[90]**	**Large test head[90]**
Maximum number of boards per system	8	16	32	32
Maximum number of channels per system (E8023CSH)	512	1024	2048	2048
Maximum number of channels per system (E8023CS)	1024	2048	2048	2048

- Parallel connection (ganging) possible for any number of successive channels within a board.
- Up to 64 ganging groups of successive channels are supported within each card.
- Maximum source/sink current per board cannot be exceeded.
- Within a test head, a DPS128 card can be mixed with any other supported device power supply.

[90] Applies to test head with DUT Scale interface and standard pogo block

- Requires E7010E or E7010F Enhanced Diagnostics Interconnect Board.
- E8023CS only: 32 boards per system only possible if 4 pogo-block version is used (2 channels ganged to 1 channel and 2.0A @2.5V per channel). For more information about the E8023PC pogo cable see TDC topic 141738.

Supply voltage / current range specifications (high current)

Voltage force	Range	Resolution	Accuracy
	-2.5 V to 7 V	200 µV	±3 mV

Voltage measure	Range	Resolution	Accuracy
	-2.5 V to 7 V	200 µV	±2 mV[91] ±1 mV[92]

The averages of dcVI and digInOut voltage measurement are multiplied by 4 automatically by SmarTest for the whole voltage range.

Current force (clamp)	Range	Resolution	Accuracy
	200 mA to 1.0 A	50 µA	-3%, +5% of range
	40 mA to 200 mA	10 µA	-3%, +5% of range
	20 mA to 100 mA	5 µA	-3%, +5% of range
	5 mA to 25 mA	1 µA	-3%, +5% of range
	2.5 mA to 12.5 mA	0.5 µA	-3%, +5% of range
	500 µA to 2.5 mA	100 nA	-3%, +5% of range
	250 µA to 1.25 mA	50 nA	-3%, +5% of range
	50 µA to 250 µA	10 nA	-3%, +5% of range
	25 µA to 125 µA	5 nA	-3%, +5% of range
	5 µA to 25 µA	1 nA	-3%, +5% of range
	2.5 µA to 12.5 µA	0.5 nA	-3%, +5% of range
Current force (n channels ganged)	n x 200 mA to n x 1.0 A	n x 50 µA	-3%, +5% of total ganged current

High accuracy current force[93]	Range	Resolution	Accuracy
	200 mA to 1.0 A	50 µA	± (3 mA +0.2 % of setting)
	40 mA to 200 mA	10 µA	± (0.6 mA +0.2 % of setting)
	20 mA to 100 mA	5 µA	± (0.6 mA +0.2 % of setting)
	5 mA to 25 mA	1 µA	± (75 µA +0.2 % of setting)

[91] *Uses 4 averaging samples internally.*
[92] *Up to 200 mA forced current, base calibration and auto calibration period 1 month*
[93] *Requires the use of a suitable API for high accuracy force current mode and at least 1 Ω total additional resistive load (incl. DUT) in the Force path*

High accuracy current force[93]	Range	Resolution	Accuracy
	2.5 mA to 12.5 mA	0.5 µA	± (75 µA +0.2 % of setting)
	500 µA to 2.5 mA	100 nA	± (7.5 µA +0.2 % of setting)
	250 µA to 1.25 mA	50 nA	± (7.5 µA +0.2 % of setting)
	50 µA to 250 µA	10 nA	± (750 nA +0.2 % of setting)
	25 µA to 125 µA	5 nA	± (750 nA +0.2 % of setting)
	5 µA to 25 µA	1 nA	± (120 nA +0.2 % of setting)
	2.5 µA to 12.5 µA	0.5 nA	± (120 nA +0.2 % of setting)
Current force (n channels ganged)	n x 200 mA to n x 1.0 A	n x 50 µA	± (n x 3mA +0.2 % of total ganged current)

Current measurement

Range	Resolution	Accuracy
± 1.0 A	50 µA	± (1 mA +0.1 % of reading)[94]
± 200 mA	10 µA	± (250 µA +0.1 % of reading)[95]
± 100 mA	5 µA	± (250 µA +0.1 % of reading)[95]
± 25 mA	1 µA	± (25 µA +0.1 % of reading)[96]
± 12.5 mA	0.5 µA	± (25 µA +0.1 % of reading)[96]
± 2.5 mA	100 nA	± (2.5 µA + 0.1 % of reading)[97]
± 1.25 mA	50 nA	± (2.5 µA + 0.1 % of reading)[98]
± 250 µA	10 nA	± (250 nA + 0.1 % of reading) [99]

[93] *Requires the use of a suitable API for high accuracy force current mode and at least 1 Ω total additional resistive load (incl. DUT) in the Force path*

[94] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 220 µF. More samples required for larger capacitances.*

[95] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 47 µF. More samples required for larger capacitances.*

[96] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 15 µF. More samples required for larger capacitances.*

[97] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.*

[98] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 2.2 µF. More samples required for larger capacitances.*

[99] *Uses 64 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.*

Range	Resolution	Accuracy
± 125 µA	5 nA	± (250 nA + 0.1 % of reading)[100]
± 25 µA	1 nA	± (50 nA + 0.2 % of reading)[100]
± 12.5 µA	0.5 nA	± (50 nA + 0.2 % of reading)[100]
n x (±1.0 A) (n channels ganged)	n x 50 µA	± (n x 1 mA +0.1 % of total current)

The averages of dcVI and digInOut current actions are automatically multiplied by SmarTest depending on the available hardware to ensure best accuracy for your measurement. This means you do not need to set higher averages than 1. For example, on a current measurement in the ± 1.0 A range setting the averages to 1, SmarTest applies 1 x 16 = 16 averages. If you set the averages to 5, SmarTest applies 5 x 16 = 80 averages.

Supported "current force" and "current measurement" range combinations

In voltage force mode, current measure range can be changed without prior disconnect.

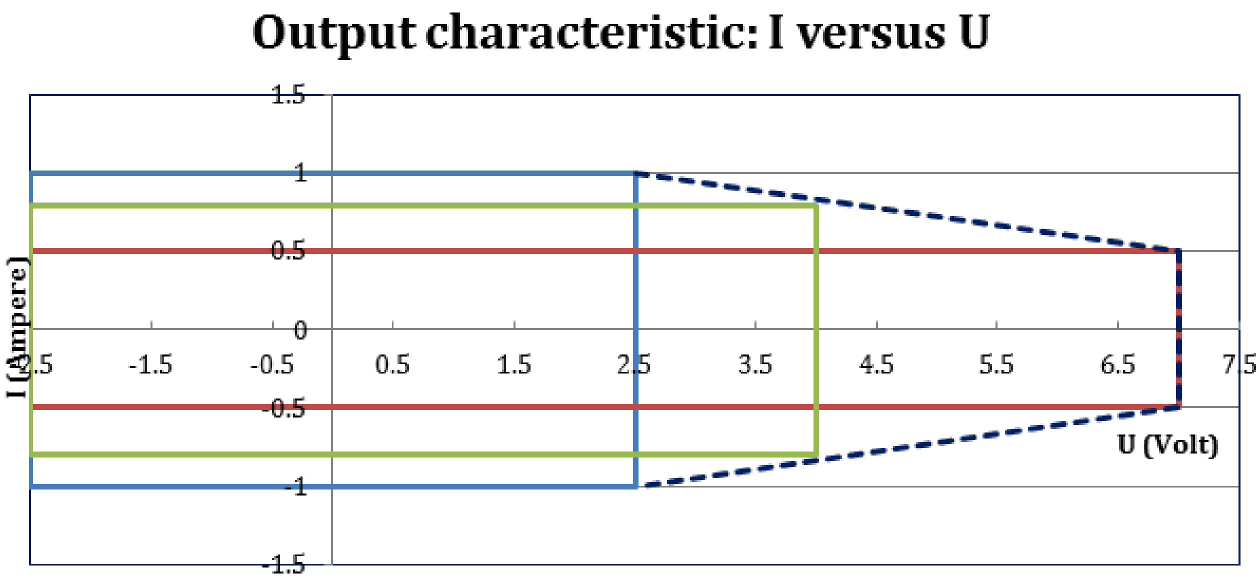
	Current force range										
Current measure range	12.5 µA	25 µA	125 µA	250 µA	1.25 mA	2.5 mA	12.5 mA	25 mA	100 mA	200 mA	1 A
±12.5 µA	ok		ok		ok		ok		ok		ok
±25 µA		ok		ok		ok		ok		ok	
±125 µA			ok		ok		ok		ok		ok
±250 µA				ok		ok		ok		ok	
±1.25 mA					ok		ok		ok		ok
±2.5 mA						ok		ok		ok	
±12.5 mA							ok		ok		ok
±25 mA								ok		ok	
±100 mA									ok		ok
±200 mA										ok	
±1 A											ok

DPS128 power diagram high current

The programmed voltage determines the maximum current. If 7 V is programmed, the maximum current is 0.5 A (red box), if 2.5 V is programmed the maximum current is 1 A for all voltages below 2.5 V (blue box).

The power diagram applies to an additional voltage drop in the Force path beyond the pogo pin of ≤ 0.1 V.

[100] *Uses 64 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 2.2 µF. More samples required for larger capacitances.*

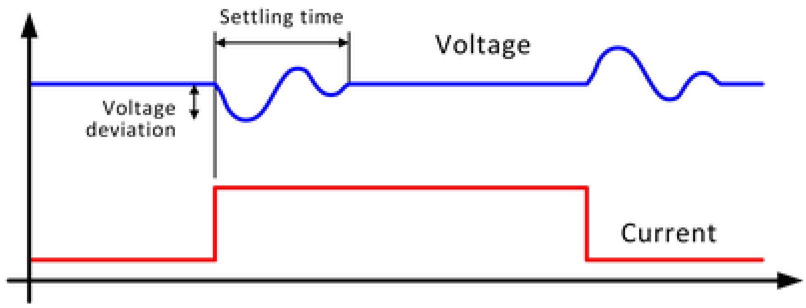


For voltages between 2.5 V and 7 V (green box), the maximum current can be calculated as follows:

$$I_{green} = \frac{I_{red} - I_{blue}}{U_{red} - U_{blue}} (U_{green} - U_{blue}) + I_{blue}$$

Dynamic behavior

Explanation of terms: Response time



Settling time after loadstep

at 2 V, settling into ±20 mV of programmed value

Loadstep	1 µF	47 µF	100 µF
0.1 A to 1.0 A	20 µs	25 µs	30 µs
1.0 A to 0.1 A	20 µs	25 µs	30 µs

Voltage deviation after loadstep

Loadstep	1 µF	47 µF	100 µF
0.1 A to 1.0 A	320 mV	120 mV	85 mV
1.0 A to 0.1 A	500 mV	125 mV	85 mV

Settling times for DPS128 measurement ranging

Wait times are to be considered when changing the DPS128 measurement range:

- Auto-range execution time: The required wait time to allow the auto-range algorithm to finish.
- Range settling time: The required wait time to achieve the specified accuracy of the target range.

For more information, refer to the "Settling times for DPS128 measurement ranging" section in topic "DPS128 / DPS64 specific setup tasks" (140640).

Other performance characteristics

Disconnect state	
Input leakage current	< 20 nA (AC relay open)

Voltmeter mode (high impedance mode)	
Voltage range	-2.5 V to +7 V
Voltage resolution	200 µV
Voltage accuracy	±1 mV[92]
Input leakage current	<30 nA

Compliance range (at disconnect state)	
Maximum external voltage at force input	-6 V to+12 V
Maximum external voltage at sense input	-7 V to +16 V

Load capacitance range	
Required external capacitance for high frequency noise filtering	1nF to 10nF[101]
Maximum supported external capacitance with stability maintained	470 µF per channel (n x 470 µF up to 5 mF if ganged)[102]
Required external capacitance range for DPS style operation	1 - 100 µF per channel[102]

Slew rate control	
Slew rate control	On-, off-, change level- rate individually controlled for each channel

DC profiling	
DC profiling	Supported per channel for voltage and current profiles
Sample rate	Up to 100 ksps

Data analysis	
Data analysis	raw data are stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board

[101] It is recommended to place a low ESL capacitance close to the pogo location. The detailed setup depends on the configuration of the DUT interface board. For more details see topic 149015 in the DUT board design reference.

[102] In addition to "Required external capacitance for high frequency noise filtering".

Data analysis	
Result analysis	built-in result analysis available that indicate the measurement data is pass or fail against a user specified test limit

Surge tracker

Voltage or current surges are detected against programmable min. and max. values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.

Voltage compare mode	
Minimum detectable pulse width	1 µs
Threshold voltage accuracy	± 10 mV typical
Input and threshold range	-2.5 V to +7 V

Current compare mode	
Minimum detectable pulse width[103]	1 µs
Threshold current accuracy	± 10 % of selected current range typical
Input and threshold range	Same as current measurement range

Analog performance

AWG mode (16 bit DAC)	
Step rate	200 ksps (kilo samples per second)
Analog bandwidth (-3 dB, 1 V swing)	5 kHz
Total channel memory	28 MB

Digitizer mode (16 bit ADC)	
Sample rate	90 ksps (kilo samples per second)
Signal path bandwidth (-3dB)	20 kHz
Total channel memory	28 MB, 11 to 22 bytes required per sample

Trigger functions	
Trigger	sequencer#controlled (no external trigger pin required)

[103] *in current range 1A*

DC Scale DPS128 Device Power Supply (E8023CSV/E8023CSW) - high voltage specifications

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Configuration

E8023CSV	High current	High voltage
Maximum number of channels per board	64	64
Maximum source/sink current per channel	1.0 A @ 2.5 V	
Maximum source/sink current per channel	0.5 A @ 7 V	0.2 A @ -6 to +15 V
Maximum source/sink current per board	62 A/54 A @ 2.5 (only high current channels used)	12.8 A/12.8 A (only high voltage channels used)
Voltage range	-2.5 V to +7 V	-6 V to +15 V

E8023CSW	High voltage
Maximum number of channels per board	128
Maximum source/sink current per channel	0.2 A @ -6 to +15 V
Maximum source/sink current per board	13.2 A/24.4 A
Voltage range	-6 V to +15 V

The DPS128/DPS64 card features two power domains with a dedicated power budget. Each domain is able to deliver half of the overall power budget. The power domain to channel mapping must be observed for DPS128 cards when assigning the available channels to the DUT pins. For more information refer to TDC topic DPS128/DPS64 power budget (147811).

	A#Class test head[104]	Compact test head[104]	Small test head[104]	Large test head[104]
Maximum number of boards per system	8	16	16	16
Maximum number of channels per system (depends on product structure decision)	1024	2048	2048	2048

[104] *Applies to test head with DUT Scale interface and standard pogo block*

- Parallel connection (ganging) possible for any number of successive channels within a board.
- Up to 64 ganging groups of successive channels are supported within each card.
- Maximum current per board cannot be exceeded.
- Sink capability per channel according to power charts.
- Within a test head, a DPS128 card can be mixed with any other device power supply.
- Requires E7010E or E7010F Enhanced Diagnostics Interconnect Board.

Supply voltage / current range specifications (high voltage)

Voltage force	Range	Resolution	Accuracy
	-6 V to 15 V	500 µV	±5 mV

Voltage measure	Range	Resolution	Accuracy
	-6 V to 15 V	100 µV	±4 mV[105]

Current force (clamp)	Range	Resolution	Accuracy
	20 to 200 mA	10 µA	-3%, +5% of range
	2.5 mA to 25 mA	1 µA	-3%, +5% of range
	250 µA to 2.5 mA	100 nA	-3%, +5% of range
	25 µA to 250 µA	10 nA	-3%, +5% of range
	2.5 µA to 25 µA	1 nA	-3%, +5% of range
Current force (n channels ganged)	n x 20 mA to n x 200 mA	n x 10 µA	-3%, +5% of total ganged current

High accuracy current force[106]	Range	Resolution	Accuracy
	20 mA to 200 mA	10 µA	± (1 mA +0.2 % of setting)
	2.5 mA to 25 mA	1 µA	± (100 µA +0.2 % of setting)
	250 µA to 2.5 mA	100 nA	± (10 µA +0.2 % of setting)
	25 µA to 25 µA	10 nA	± (1 µA +0.2 % of setting)
	2.5 µA to 25 µA	1 nA	± (100 nA +0.2 % of setting)
Current force (n channels ganged)	n x 200 mA to n x 1.0 A	n x 10 µA	± (n x 1mA +0.2 % of total ganged current)

Current measurement

Range	Resolution	Accuracy
± 200 mA	10 µA	± (300 µA + 0.1% of reading)[107]

[105] Uses 4 averaging samples internally.
[106] Requires the use of a suitable API for high accuracy force current mode and at least 1 Ω total additional resistive load (incl. DUT) in the Force path
[107] Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 47 µF. More samples required for larger capacitances.

Range	Resolution	Accuracy
± 25 mA	1 µA	± (30 µA + 0.1% of reading)[108]
± 2.5 mA	100 nA	± (3 µA + 0.1% of reading)[109]
± 250 µA	10 nA	± (300 nA + 0.1% of reading)[110]
± 25 µA	1 nA	± (60 nA + 0.1% of reading)[111]
n x (± 200 mA) (n channels ganged)	n x 10 µA	± (n x 300 µA +0.1% of total current)

The averages of dcVI and digInOut current actions are automatically multiplied by SmarTest depending on the available hardware to ensure best accuracy for your measurement. This means you do not need to set higher averages than 1. For example, on a current measurement in the ± 1.0 A range setting the averages to 1, SmarTest applies 1 x 16 = 16 averages. If you set the averages to 5, SmarTest applies 5 x 16 = 80 averages.

Supported current force and current measurement range combinations

In voltage force mode, current measure range can be changed without prior disconnect.

		Current force range				
Current measure range		25 µA	250 µA	2.5 mA	25 mA	200 mA
	± 25 µA	ok	ok	ok	ok	ok
	± 250 µA		ok	ok	ok	ok
	± 2.5 mA			ok	ok	ok
	± 25 mA				ok	ok
	± 200 mA					ok

DPS128 power diagram high voltage

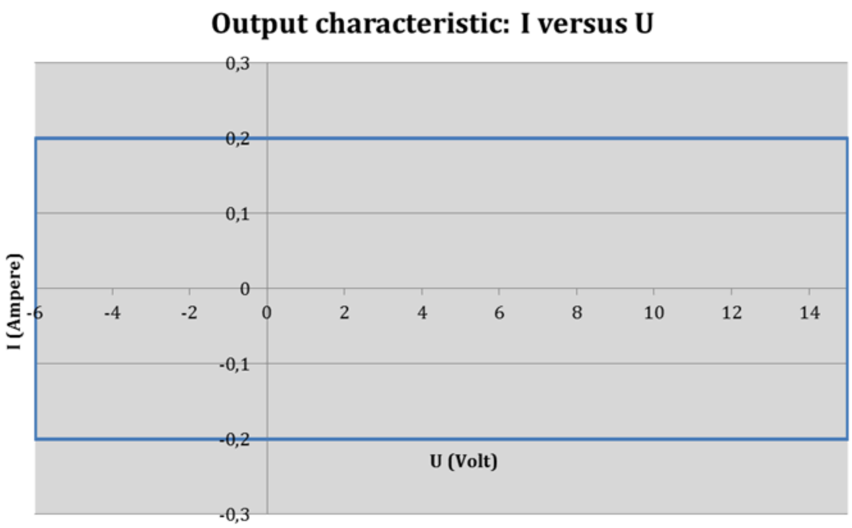
The power diagram applies to an additional voltage drop in the Force path beyond the pogo pin of = 0.1V.

[108] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 15 µF. More samples required for larger capacitances.*

[109] *Uses 16 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.*

[110] *Uses 64 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.*

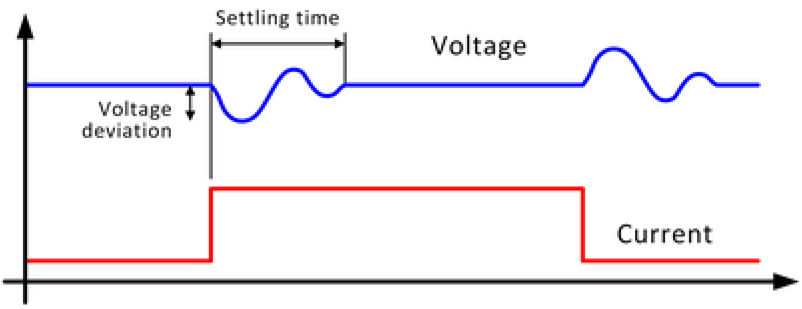
[111] *Uses 64 averaging samples internally. Maximum supported external capacitance to achieve accuracy is 2.2 µF. More samples required for larger capacitances. Applies to = 50% relative humidity.*



DPS128 power diagram high voltage

Dynamic behavior

Explanation of terms: Response time



Settling time after loadstep

at 5 V, settling into ±20 mV of programmed value

Loadstep	100 nF	1 µF
0.1 A to 0.2 A	12 µs	12 µs
0.2 A to 0.1 A	12 µs	12 µs

Voltage deviation after loadstep

Loadstep	100 nF	1 µF
0.1 A to 0.2 A	190 mV	100 mV
0.2 A to 0.1 A	200 mV	110 mV

Settling times for DPS128 measurement ranging

Wait times are to be considered when changing the DPS128 measurement range:

- Auto-range execution time: The required wait time to allow the auto-range algorithm to finish.
- Range settling time: The required wait time to achieve the specified accuracy of the target range.

For more information, refer to the "Settling times for DPS128 measurement ranging" section in topic "DPS128 / DPS64 specific setup tasks" (140640).

Other performance characteristics

Disconnect state	
Input leakage current	< 40 nA (AC relay open)

Compliance range (at disconnect state)	
Maximum external voltage at force input	-11 V to +17 V
Maximum external voltage at sense input	-11 V to +17 V

Load capacitance range	
Required external capacitance for high frequency noise filtering	1nF to 10nF[112]
Maximum supported external capacitance with stability maintained	22 µF per channel (47 µF total if ganged)[113]
Required external capacitance range for DPS style operation	100 nF - 4.7 µF per channel[113]

Slew rate control	
Slew rate control	On-, off-, change level- rate individually controlled for each channel

DC profiling	
DC profiling	Supported per channel for voltage and current profiles
Sample rate	Up to 100 ksps

Data analysis	
Data analysis	raw data are stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board
Result analysis	built-in result analysis available that indicate the measurement data is pass or fail against a user specified test limit

Voltmeter mode (high impedance mode)	
Voltage range	-6 V to +15 V
Voltage resolution	500 µV
Voltage accuracy	± 4 mV
Voltage accuracy	± 3 mV[114]
Input leakage current	< 40 nA

[112] It is recommended to place a low ESL capacitance close to the pogo location. The detailed setup depends on the configuration of the DUT interface board. For more details see topic 149015 in the DUT Board Design Guide.
[113] In addition to "Required external capacitance for high frequency noise filtering".
[114] Base calibration and auto calibration period 1 month

Analog performance

AWG mode (16 bit DAC)	
Step rate	200 ksps (kilo samples per second)
Analog bandwidth (-3 dB, 1 V swing)	5 kHz
Total channel memory	28 MB

Digitizer mode (18 bit ADC)	
Sample rate	90 ksps (kilo samples per second)
Signal path bandwidth (-3dB)	20 kHz
Total channel memory	28 MB, 11 to 22 bytes required per sample

Trigger functions	
Trigger	sequencer#controlled (no external trigger pin required)

DC Scale FVI16 Floating Power VI Source (E8026PV)

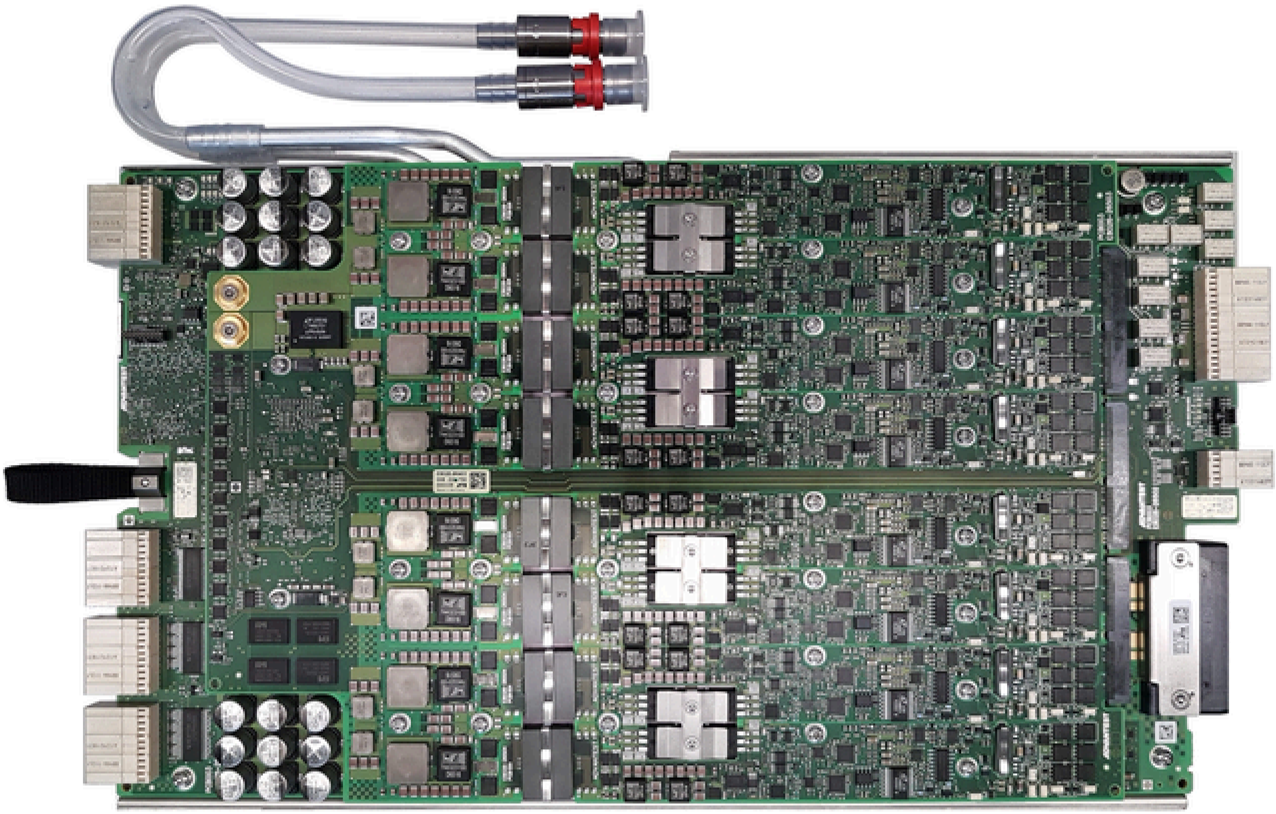
The FVI16 card is a high-current and high-voltage floating VI source that is optimized for a wide range of analog power tests. The card belongs to the DC Scale family and can reside in any channel card slot.

FVI16 provides 16 fully floating high-current and high-voltage channels. Each channel can supply voltages from -60 V to +120 V within a floating range of ±200 V and offers pulse current capability up to ±10 A per channel and ±3 A continuous.

A floating differential high-voltage time measurement unit (TMU) with separate inputs is available for each channel.

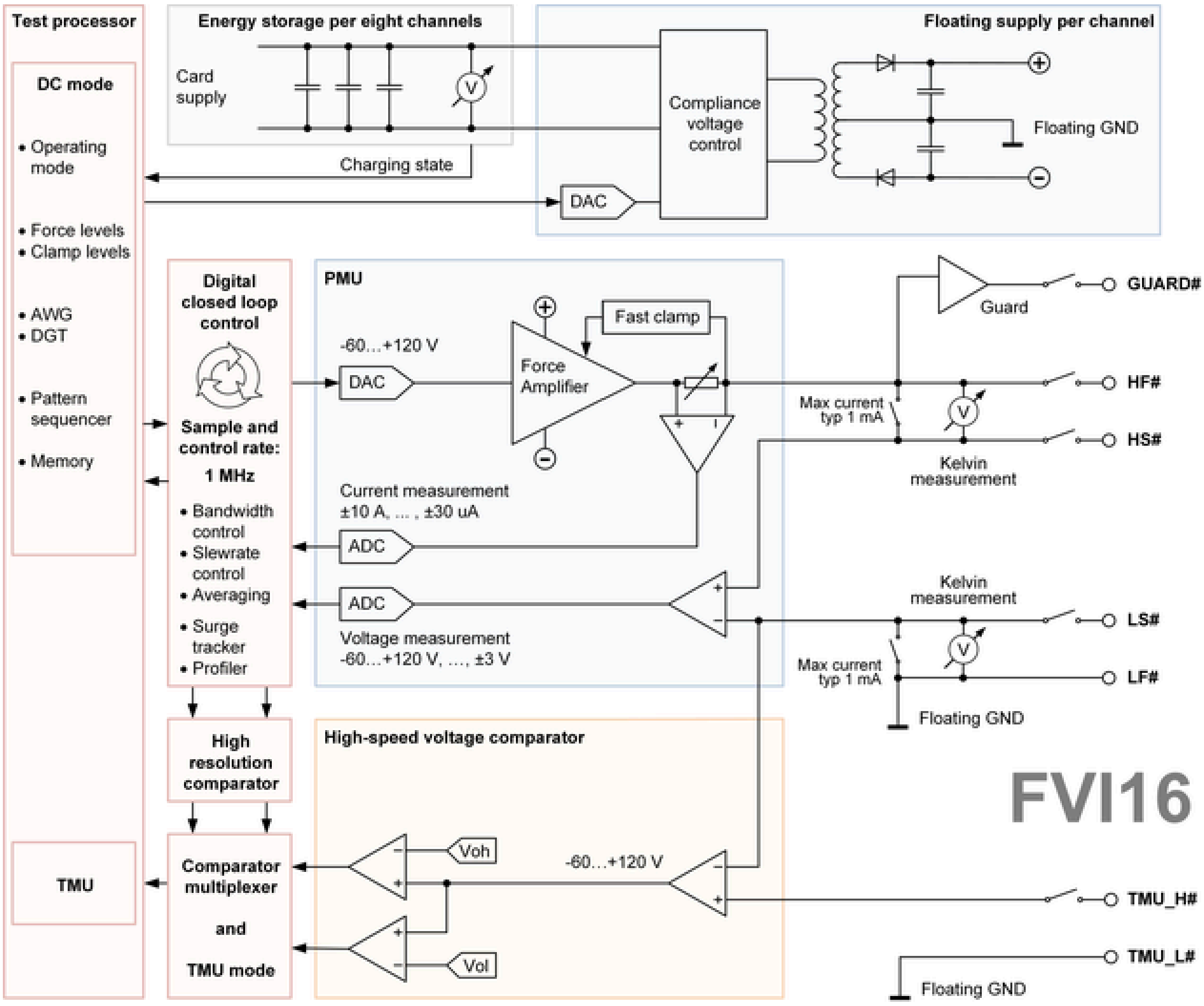
Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.



FVI16 block diagram and configuration

Block diagram



Configuration

	FVI16			
Number of VI channels per card	16 (2 groups, 8 channels each)			
	A -Class	C-Class	S-Class	L-Class
Maximum number of cards per system	8	16	32	64
Maximum Power VI channels per system	128	256	512	1024

Other configuration data

- FVI16 is slot compatible to AVI64, DPS64, DPS128, and Pure Clock cards.
- All channels are fully floating.
- All channels have guard lines.
- FVI16 card can be mixed with any other device power supply, except PVI8.
- Diagnostics requires the Diagnostics Interconnect Board E8026DIAG to cover interconnection tests.

FVI16 voltage force and measurement specifications

Voltage force	Range	Resolution	Accuracy
High-power mode	± 3 V	> 17 bit	± (0.2 mV + 0.02 % of setting) ± (40 µV + 0.003 % of relative setting)[115]
	± 10 V	> 17 bit	± (1 mV + 0.02 % of setting) ± (80 µV + 0.003 % of relative setting)[115]
	± 30 V	> 17 bit	± (3m V + 0.02 % of setting) ± (700 µV + 0.01 % of relative setting)[115]
	± 60 V	> 17 bit	± (8 mV + 0.02 % of setting) ± (1 mV + 0.01 % of relative setting)[115]
Programmable compliance voltage	+10 V ... +65 \	> 12 bit	
High-voltage mode (max. 30 mA DC)	-60 V ... +120	> 17 bit	± (20 mV + 0.03 % of setting) ± (3.5 mV + 0.01 % of relative setting)[115]

Stacked voltage force accuracy: N times single channel voltage accuracy

Voltage measurement	Range[116]	Resolution	Accuracy[117]
High-power mode	± 3 V	25 µV	± (0.2 mV + 0.02 % of reading) ± (40 µV + 0.003 % of relative reading)[118]
	± 10 V	80 µV	± (1 mV + 0.02 % of reading) ± (80 µV + 0.003 % of relative reading)[118]
	± 30 V	250 µV	± (3 mV + 0.02 % of reading) ± (700 µV + 0.01 % of relative reading)[118]
	± 60 V	800 µV	± (8 mV + 0.02 % of reading) ± (1 mV + 0.01 % of relative reading)[118]
High-voltage mode (max. 30 mA DC)	-60 V ... +120	2 mV	± (20 mV + 0.03 % of reading) ± (3.5 mV + 0.01 % of relative reading)[118]

Stacked voltage measurement accuracy: N times single channel voltage accuracy

[115] The relative accuracy specification describes the setting repeatability of a channel within 6 hours.
[116] Voltage measurement range is controlled by voltage force settings (or voltage clamp, if active).
[117] Requires averaging over 500 samples and 20 kHz measurement filter.
[118] The relative accuracy specification describes the repeatability of measurements of a channel within 6 hours.

Current clamp	Range	Resolution	Accuracy[119]
High-power mode	±10 A	> 15 bit	± (10 mA + 0.2 % of setting)
	±3 A	> 15 bit	± (2.5 mA + 0.2 % of setting)
	±300 mA	> 15 bit	± (250 µA + 0.2 % of setting)
High-power mode or High-voltage mode	±30 mA	> 15 bit	± (25 µA + 0.2 % of setting)
	±3 mA	> 15 bit	± (2.5 µA + 0.2 % of setting + 50 nA/V * $\lvert V_{act}\rvert$)
	±300 µA	> 15 bit	± (250 nA + 0.2 % of setting + 5 nA/V * $\lvert V_{act}\rvert$)
	±30 µA	> 15 bit	± (25 nA + 0.2 % of setting + 2 nA/V * $\lvert V_{act}\rvert$)

Ganged current clamp accuracy: N times single channel clamp accuracy

[119] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the FVI16 channel is in (forcing or clamping).*

FVI16 current force and measurement specifications

Current force	Range[120]	Resolution	Accuracy[121]
High-power mode	±10 A	> 16 bit	± (2 mA + 0.25 % of setting + 0.1 mA/V * |V_{act}|)
	±3 A	> 16 bit	± (1.5 mA + 0.15 % of setting + 33 µA/V * |V_{act}|)
	±300 mA	> 17 bit	± (150 µA + 0.05 % of setting + 5 µA/V * |V_{act}|)
High-power mode or High-voltage mode	±30 mA	> 17 bit	± (20 µA + 0.05 % of setting + 0.5 µA/V * |V_{act}|)
	±3 mA	> 17 bit	± (500 nA + 0.05 % of setting + 50 nA/V * |V_{act}|)
	±300 µA	> 17 bit	± (50 nA + 0.05 % of setting + 5 nA/V * |V_{act}|)
	±30 µA	> 17 bit	± (10 nA + 0.1 % of setting + 1 nA/V * |V_{act}|)

Ganged current force accuracy: N times single channel current force accuracy

Current measurement	Range[122]	Resolution	Accuracy[121][123]
High-power mode	±10 A	> 16 bit	± (2 mA + 0.25 % of reading + 0.1 mA/V * |V_{act}|)
	±3 A	> 16 bit	± (1.5 mA + 0.15 % of reading + 33 µA/V * |V_{act}|)
	±300 mA	> 17 bit	± (150 µA + 0.05 % of reading + 5 µA/V * |V_{act}|)
High-power mode or High-voltage mode	±30 mA	> 17 bit	± (20 µA + 0.05 % of reading + 0.5 µA/V * |Vact|)
	±3 mA	> 17 bit	± (500 nA + 0.05 % of reading + 50 nA/V * |V_{act}|)
	±300 µA	> 17 bit	± (50 nA + 0.05 % of reading + 5 nA/V * |V_{act}|)
	±30 µA	> 17 bit	± (10 nA + 0.1 % of reading + 1 nA/V * |V_{act}|)

Ganged current force accuracy: N times single channel current force accuracy

Voltage clamp	Range	Resolution	Accuracy
High-power mode	± 3 V	> 17 bit	± (3 mV + 0.1 % of setting)
	± 10 V	> 17 bit	± (10 mV + 0.1 % of setting)
	± 30 V	> 17 bit	± (30 mV + 0.1 % of setting)
	± 60 V	> 17 bit	± (60 mV + 0.1 % of setting)
High-voltage mode	-60 V ... +120	> 17 bit	± (120 mV + 0.1 % of setting)

[120] Current ranges can be selected independent of programmed voltage range.
[121] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the FVI16 channel is in (forcing or clamping).
[122] Current measurement range is controlled by current force settings (or current clamp, if active).
[123] Requires averaging over 500 samples and 20 kHz measurement filter.

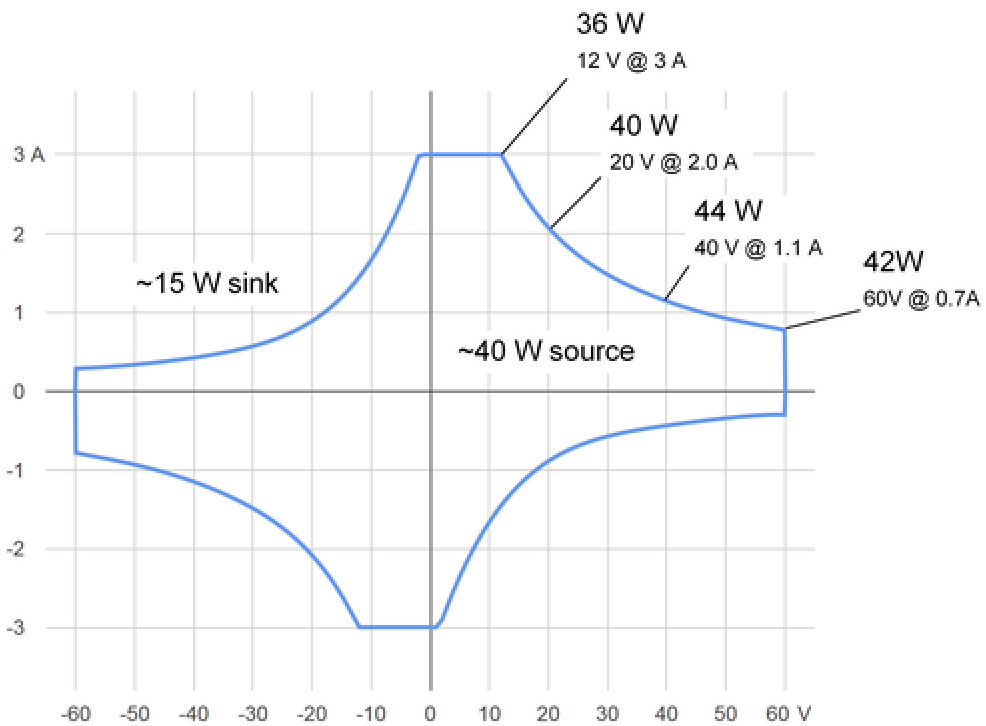
FVI16 voltage and current capability specifications

All specifications and diagrams in this section are valid at the pogo pins.

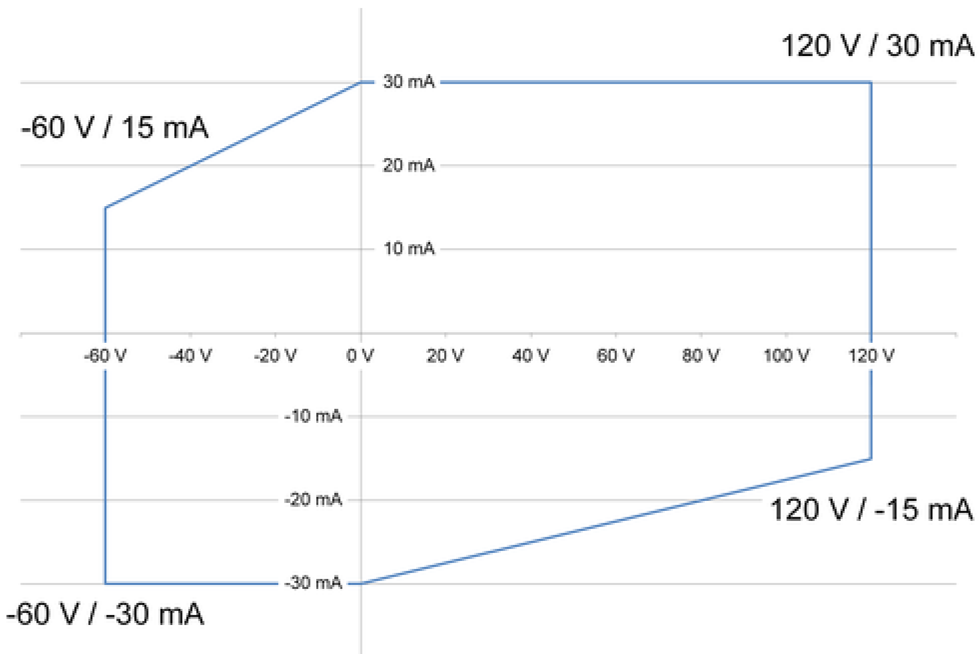
Continuous voltage and current capabilities per channel

The maximum total power budget per card is 400 W. This budget is equally distributed to two channel groups 1 … 8 and 9 … 16 providing 200 W per group.

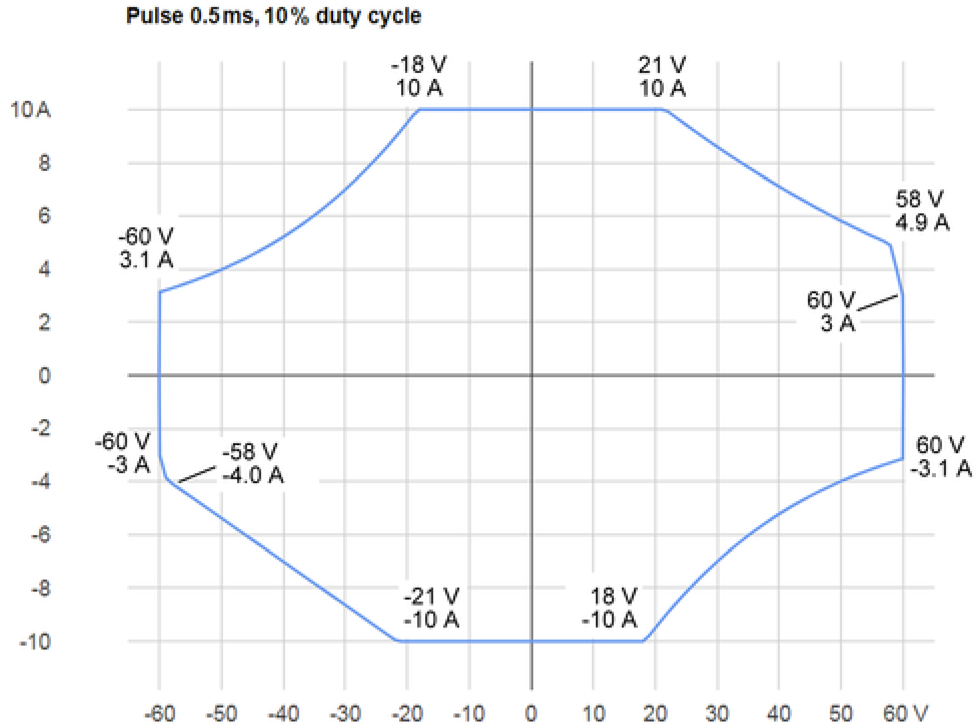
High-power mode:



High-voltage mode:



Maximum pulse power per channel (~250W)

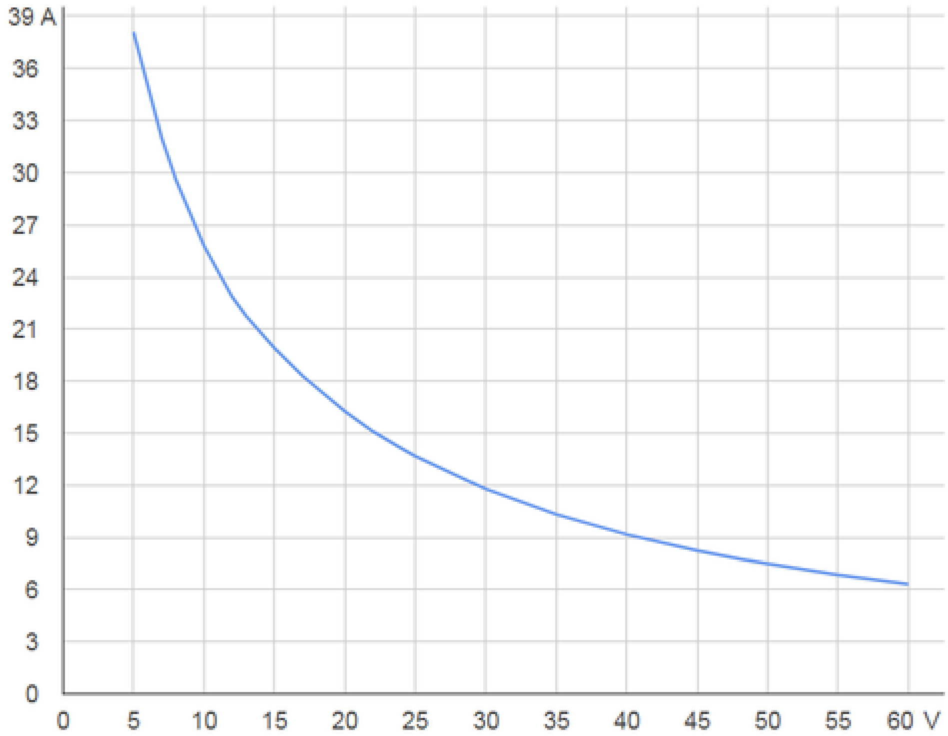


The diagram above shows the pulse power rating in high-power mode for 0.5 ms pulse width and 10 % duty cycle:

- X-axis: Voltage in [V]
- Y-axis: Current in [A]

Typical maximum continuous voltage and current capability per card

The following diagram shows the typical maximum continuous voltage and current capability per card at pogo interface with all 16 channels ganged.


39 A
36
33
30
27
24
21
18
15
12
9
6
3
0
0
5
10
15
20
25
30
35
40
45
50
55
60 V

FVI16 digitizer specifications

Per-pin voltage digitizer[124]	Range	Resolution	Accuracy[125]	INL typ.
High-power mode	±3 V	25 µV	± (1 mV + 0.02 % of reading)	± 3 LSB
	±10 V	80 µV	± (5 mV + 0.02 % of reading)	± 3 LSB
	±30 V	250 µV	± (15 mV + 0.02 % of reading)	± 6 LSB
	±60 V	800 µV	± (20 mV + 0.02 % of reading)	± 6 LSB
High-voltage mode	-60 V … +120V	2 mV	± (40 mV + 0.03 % of reading)	± 8 LSB
Input impedance (typical)	> 1 GΩ			
Input capacitance (typical)	200 pF			
Sampling rate	> 0 … 1 Msps (Million samples per second, ADC sample clock 1 MHz)			
Analog bandwidth (typical, -3 dB)	250 kHz			
Programmable measurement filter	100 Hz … 250 kHz			
Waveform capture memory per pin	not exceeding total available memory size per channel (107 MByte)			

Per-pin current digitizer[124]

Mode	Measurement filter	Range	Resolution	Accuracy[126]
High-power	1 kHz	±30 µA	> 17 bit	± (50 nA + 0.1 % of reading + 2 nA/V * | V_{act} |)
High-voltage				± (100 nA + 0.1 % of reading + 2 nA/V * | V_{act} |)
High-power	1 kHz	±300 µA	> 17 bit	± (250 nA + 0.05 % of reading + 5 nA/V * | V_{act} |)
High-voltage				
High-power	1 kHz	±3 mA	> 17 bit	± (2.5 µA + 0.05 % of reading + 50 nA/V * | V_{act} |)
High-voltage				
High-power	20 kHz	±30 mA	> 17 bit	± (100 µA + 0.05 % of reading)
High-voltage				
High-power	20 kHz	±300 mA	> 17 bit	± (750 µA + 0.05 % of reading)
		±3 A	> 17 bit	± (7.5 mA + 0.15 % of reading)
		±10 A	> 16 bit	± (15 mA + 0.25 % of reading)
Sampling rate		> 0 … 1 Msps (Million samples per second, ADC sample clock 1MHz)		
Programmable measurement filter		100 Hz … 250 kHz		
Waveform capture memory per pin		not exceeding total available memory size per channel (107 MByte)		

[124] *Voltage and current digitizer are available for every channel and can be used in parallel. Software selectable range settings are the same as for VI modes.*

[125] *Condition: Single sample and measurement filter set to 20 kHz*

[126] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the FVI16 channel is in (forcing or clamping).*

FVI16 AWG specifications

The FVI16 output can be sequencer#controlled via events (defined in pattern, stored in test processor memory) to generate arbitrary waveforms.

Per-pin voltage AWG		
Voltage ranges		±3 V, ±10 V, ±30 V, ±60 V, -60 V ... +120 V
Resolution		> 17 bit
Accuracy		See voltage force accuracy of ranges (after settling)
Maximum current		See voltage / current capabilities (VI ratings)
Sampling rate		> 0 ... 1 Msps (Million samples per second)
Analog bandwidth (typical)		50 kHz
Waveform memory (per pin)[127]		Up to 1 Msamples (Million samples)
INL (typical)	±3 V range:	±3 LSB
	±10 V range:	±3 LSB
	±30 V range:	±6 LSB
	±60 V range:	±6 LSB
	-60 V ... +120 V range:	±8 LSB

Per-pin current AWG	
Current ranges	±30 µA, ±300 µA, ±3 mA, ±30 mA, ±300 mA, ±3 A, ±10 A
Resolution	> 17 bit
Accuracy	See current force accuracy of ranges (after settling)
Maximum voltage	See voltage / current capabilities (VI ratings)
Sampling rate	> 0 ... 1 Msps (Million samples per second)
Analog bandwidth (typical)	50 kHz
Waveform memory (per pin)[127]	Up to 1 Msamples (Million samples)

[127] Waveform memory shared between all AWG modes and DC test points.

FVI16 TMU specifications

Timing Measurement Unit (TMU)

Timing measurements are done using the comparator functionality available per channel (see comparator specifications above).

The TMU allows for time delay / skew measurements (V_{OH} / I_{CH} # V_{OH} / I_{CH}, V_{OL} / I_{CL} # V_{OL} / I_{CL}) also across different instrument types (i.e. PinScale1600), frequency / pulse width measurements, and rise / fall time measurements (V_{OL} / I_{CL} # V_{OH} / I_{CH}, V_{OH} / I_{CH} # V_{OL} / I_{CL}).

Timing Measurement Unit (TMU)[128]	
DC / AC Performance	See comparators
Accuracy for rise / fall time measurements (single shot)	±4 ns
Accuracy for skew to other FVI16 TMU channel (single shot)	±4 ns
Accuracy for single time stamp to ARM (time zero)	±2 ns

The TMU can be used with all compare modes as defined below. The following combinations are supported.

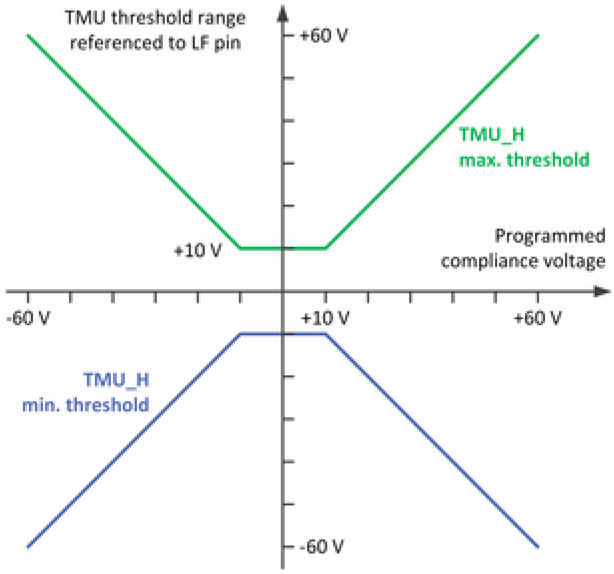
Compare modes	VI Comparator	Inputs
High-speed voltage	n.a.	TMU_H# and TMU_L#
High-resolution voltage	supported	HS# and LS#
High-resolution current	supported	HF#

High-speed (HS) comparator inputs	
Programmable thresholds	V_{OL}, V_{OH}
Input voltage and threshold ranges	-60 V ≤ V_{OL}, V_{OH} ≤ +120 V
Threshold accuracy	± (200 mV + 0.1 % of setting)
TMU_H safe input voltage	Clamped to programmed VI output compliance voltage of same channel. Input voltage max 20 V over programmed compliance range of same channel.
TMU_L safe input voltage	Keep at the same voltage level as LS pin of the channel
Threshold hysteresis	25 mV (typical)
Programmable compare event hold-off time	0.0 … 20.475 µs Default hold-off time: 100 ns
Input impedance (typical)	>1 MΩ
Signal path bandwidth	45 MHz (typical)
Compare timing resolution	1 ps
PMU dependency	See diagram below

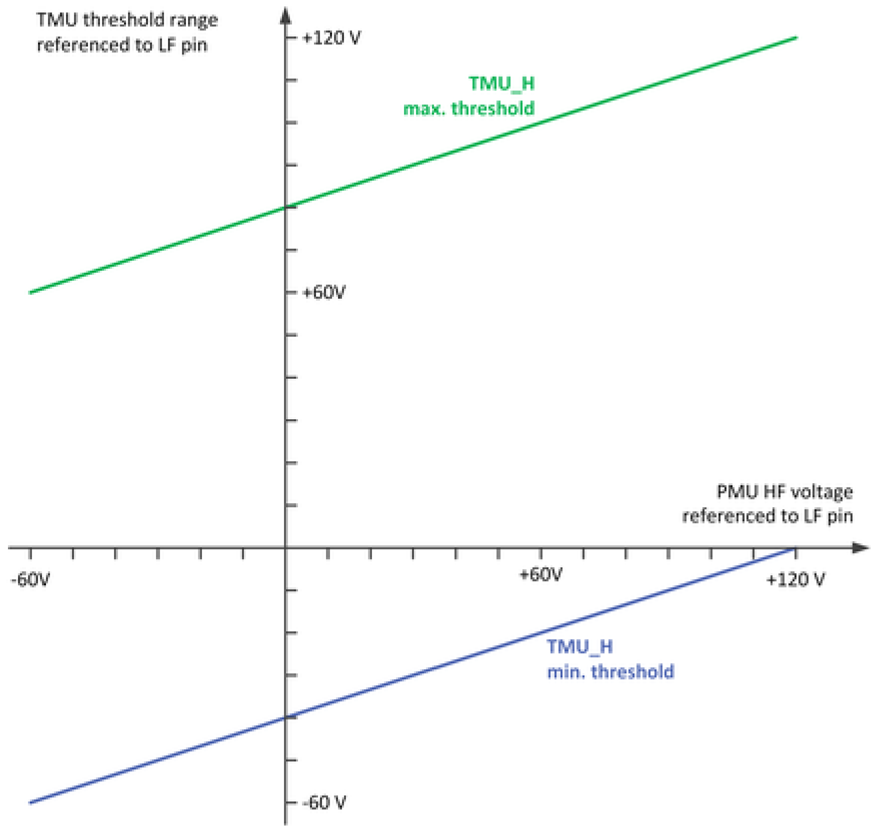
High-speed (HS) comparator threshold range

High-power (±60 V) mode:

[128] *Condition: Compliance voltage set to 65V*



High-voltage (-60 ... 120 V) mode:



High-resolution TMU mode (Voltage comparator via HS/LS and current comparator via HF)		
Programmable dual thresholds		V_{OL}, V_{OH}, I_{CL}, I_{CH} (with programmable hysteresis)
Threshold ranges (voltage)[129]		±3 V, ±10 V, ±30 V, ±60 V, -60 V ... +120 V
Threshold accuracy (voltage)		
	± 3 V range:	± (2 mV + 0.02 % of setting)
	± 10 V range:	± (5 mV + 0.02 % of setting)

[129] *Voltage measure range is used in compare mode.*

High-resolution TMU mode (Voltage comparator via HS/LS and current comparator via HF)		
	± 30 V range:	± (15 mV + 0.02 % of setting)
	± 60 V range:	± (40 mV + 0.02 % of setting)
	-60 V … +120 V range:	± (100 mV + 0.03 % of setting) @ 250 kHz
Threshold hysteresis (voltage)		Programmable within full range
Threshold ranges (current)[130]		±30 μA, ±300 μA, ±3 mA, ±30 mA, ±300 mA, ±3 A, ±10 A
Threshold accuracy (current)[131]		
High-current mode	± 30 μA range:	± (300 nA + 0.1 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
High-voltage mode		± (1.5 μA + 0.1 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
High-current mode	± 300 μA range:	± (3 μA + 0.05 % of setting + 5 nA/V * $\lvert V_{act} \rvert$)
High-voltage mode		± (15 μA + 0.05 % of setting + 5 nA/V * $\lvert V_{act} \rvert$)
High-current mode	± 3 mA range:	± (20 μA + 0.05 % of setting + 50 nA/V * $\lvert V_{act} \rvert$)
High-voltage mode		± (100 μA + 0.05 % of setting + 50 nA/V * $\lvert V_{act} \rvert$)
High-current mode	± 30 mA range:	± (100 μA + 0.05 % of setting + 0.5 μA/V * $\lvert V_{act} \rvert$)
High-voltage mode		± (200 μA + 0.05 % of setting + 0.5 μA/V * $\lvert V_{act} \rvert$)
High-current mode	± 300 mA range:	± (1 mA + 0.05 % of setting + 5 μA/V * $\lvert V_{act} \rvert$)
	± 3 A range:	± (10 mA + 0.15 % of setting + 33 μA/V * $\lvert V_{act} \rvert$)
	± 10 A range:	± (30 mA + 0.25 % of setting + 0.1 mA/V * $\lvert V_{act} \rvert$)
Threshold hysteresis (current)		Programmable within full range
Compare timing accuracy		± 500 ns[132]
Compare timing resolution		5 ns (interpolated)[133]
Signal path bandwidth		250 kHz (typical)

High-speed TMU mode **Typical apps cases**	**Range**	**Resolution**	**Accuracy (typ.)[134]**
Rise time (single shot measurement)	20 ns ... 100 ms	< 1 ns	±4 ns
Fall time (single shot measurement)	20 ns ... 100 ms	< 1 ns	±4 ns
Rise time (repetitive)	0 ns ... 100 ms	< 1 ns	±4 ns
Fall time (repetitive)	0 ns ... 100 ms	< 1 ns	±4 ns
Pulse-width (single shot)	20 ns ... 100 ms	< 1 ns	±4 ns
Pulse-width (time delta of repetitive pulse)	0 ns ... 100 ms	< 1 ns	±4 ns
Frequency (= multiple period duration measure)	0 ... 50 MHz	< 1 ns	±4 ns
Delay to digital	0ns ... 100 ms	< 1 ns	±4 ns[135]

[130] *Current measure range is used in compare mode.*
[131] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).*
[132] *Additional fixed channel delay in high resolution mode of 3 μs.*
[133] *Sampling rate = 1 μs*
[134] *Condition: Slew rate < tracking slew rate*
[135] *Requires application specific calibration of the path*

High-speed TMU mode **Typical apps cases**	**Range**	**Resolution**	**Accuracy (typ.)[134]**
Delay to other FVI16 channel	0ns … 100 ms	< 1 ns	±4 ns[135]

[134] *Condition: Slew rate < tracking slew rate*

FVI16 channel stacking and ganging specifications

Channel stacking	
Maximum channel floating range	±200 V to system ground[136]
Maximum stacked channels	3 channels; any channel combination within same card
Typical source stacks	+200 V: AVI64 + FVI16 = 80 V + 120 V = 200 V (applicative) +180 V: 3 x FVI16 = 3 x 60 V = 180 V (native, SW support) -180 V: 3 x FVI16 = 3 x -60 V = -180 V (native, SW support)
Native SW supported voltage ranges for stacking	±60 V
Native SW supported current ranges for stacking	±30 µA, ±300 µA, ±3 mA, ±30 mA, ±300 mA, ±3 A and ±10 A
Software support	Stacked setup represented as single channel Flexible stack / unstack within same test program
Hardware support	Routing on DUT board Layout need to have sufficient high voltage creepage distance

Channel ganging	
Maximum ganged source channels	16 channels per card with integrated software support; any channel combination within same card > 16 channels, user need to take care of individual source handling
Maximum ganged current	±155 A pulsed per card for full 16 channel setup, ±97 A pulsed per card for a 10 channel setup
Native SW supported voltage ranges for ganging	±3 V, ±10 V, ±30 V and ±60 V
Native SW supported current ranges for ganging	±10 A
Software support	Ganged setup represented as single channel Flexible gang / ungang within same test program
Hardware support	Wiring of ganged channels on DUT board; layout needs to have sufficient current capability

[136] *The voltage at any of the channel pins (HF#, HS#, LF#, LS#, TMU_H#, TMU_L#) must not exceed the floating range.*

FVI16 VI slew rate and bandwidth control specifications

Digital feedback loop enables flexible adjustments of slew rate and bandwidth of force signals to optimize the adaption upon different load conditions.

Programmable slew rate for voltage

Mode	Range	Voltage slew rate		
		Min	**Max**	**Default**
High-power	± 3 V	63 µV/µs	1 V/µs	0.5 V/µs
	± 10 V	210 µV/µs	1 V/µs	0.5 V/µs
	± 30 V	625 µV/µs	1 V/µs	0.5 V/µs
	± 60 V	2.1 mV/µs	1 V/µs	0.5 V/µs
High-voltage	-60 V ... +120 V	2.4 mV/µs	0.2 V/µs	0.1 V/µs

Programmable slew rate for current

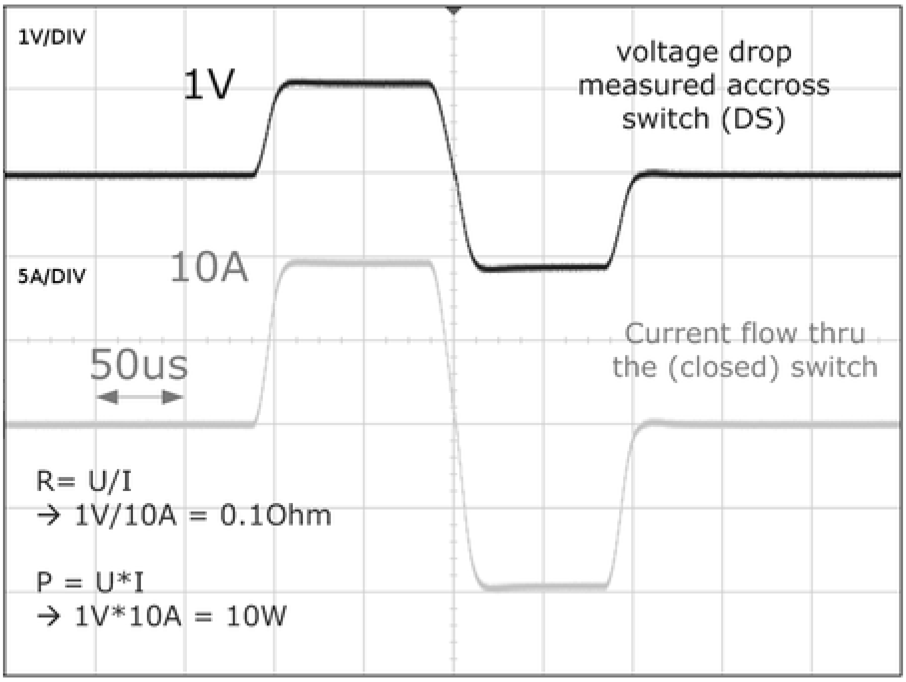
Mode	Range	Current slew rate		
		Min	**Max**	**Default**
High-power	±30 µA	540 pA/µs	3 µA/µs	3 µA/µs
	±300 µA	5.4 nA/µs	30 µA/µs	30 µA/µs
	±3 mA	54 nA/µs	300 µA/µs	300 µA/µs
	±30 mA	540 nA/µs	3 mA/µs	3 mA/µs
	±300 mA	6.9 µA/µs	30 mA/µs	30 mA/µs
	±3 A	94 µA/µs	300 mA/µs	300 mA/µs
	±10 A	517 µA/µs	1 A/µs	1 A/µs
High-voltage	±30 µA	540 pA/µs	3 µA/µs	3 µA/µs
	±300 µA	5.4 nA/µs	30 µA/µs	30 µA/µs
	±3 mA	54 nA/µs	300 µA/µs	300 µA/µs
	±30 mA	540 nA/µs	3 mA/µs	3 mA/µs

Programmable bandwidth for voltage and current

Mode	Range	Voltage filter		
		Min	**Max**	**Default**
High-power	Up to ±60 V	100 Hz	50 kHz	10 kHz
High-voltage	-60 V ... +120 V	100 Hz	5 kHz	5 kHz

Mode	Range	Current filter		
		Min	**Max**	**Default**
High-power	Up to ±60 V All current ranges	100 Hz	50 kHz	10 kHz
High-voltage	-60 V ... +120 V All current ranges	100 Hz	50 kHz	50 kHz

Typical settling times (for example, 10 A High-power RDSON Test)



Condition

R_{DUT}:	0.1 Ω
I_{DUT}:	10 A
Force mode:	CFVM
Sequence:	**1.** Pulse to slew up **2.** Settling to spec **3.** Measure voltage

Expectation

Pulse time including settling time:	< 50 µs
Stable measurement at:	< 40 µs
Maximum overshoot:	< 5 %

FVI16 load capacitance and measurement filter specifications

The digital feedback loop offers flexible adjustments to the available load capacity and offers also programmable measurement filters.

Programmable load capacitance

Mode	Range	Load capacitance		
		Min	Max	Default
High-power	±30 µA	0 nF	0.66 nF	0 nF
	±300 µA	0 nF	6.6 nF	0 nF
	±3 mA	0 nF	66 nF	0 nF
	±30 mA	0 nF	0.66 µF	0 nF
	±300 mA	0 nF	6.6 µF	0 nF
	±3 A	0 nF	33 µF	0 nF
	±10 A	0 nF	100 µF	0 nF
High-voltage	±30 µA	0 nF	0.66 nF	0 nF
	±300 µA	0 nF	6.6 nF	0 nF
	±3 mA	0 nF	66 nF	0 nF
	±30 mA	0 nF	0.66 µF	0 nF

Programmable measurement filter for voltage and current

Mode	Range	Measurement filter		
		Min	Max	Default
High-power	All voltage ranges and all current ranges	100 Hz	250 kHz	250 kHz
High-voltage	All current ranges	100 Hz	250 kHz	250 kHz

One common filter setting applies for both voltage and current measurement.

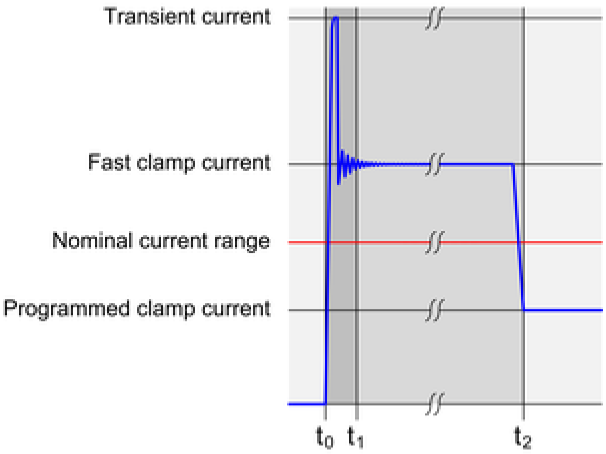
Typical measurement filter settings

Filter	Tau	Settling time (6 Tau)	Application
250 kHz	0.64 µs	3.8 µs	for fastest measurement results
50 kHz	3.18 µs	19.1 µs	for fast measurement results
20 kHz	7.96 µs	47.8 µs	for accurate measurements results
5 kHz	31.8 µs	191.0 µs	for measurements in the high-voltage mode

FVI16 fast current clamp specifications

A sudden short between HF# and LF# pins causes high current flow exceeding current limit settings for a limited time: Current spike transients.

The fast current clamp is a circuitry within the FVI16 channel to minimize the current spike energy by reducing the current and by reducing the spike duration. In a second stage the control loop reduces current flow to the programmed current clamp settings.

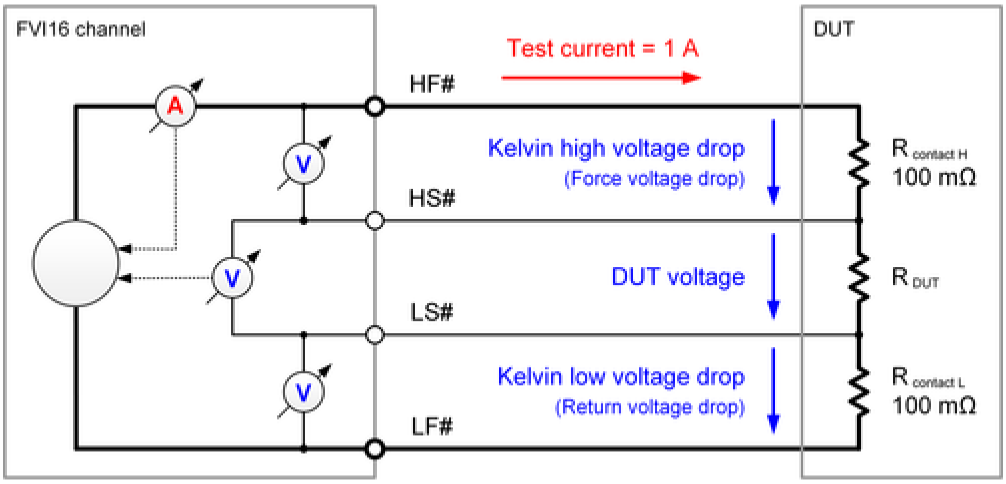


High-power mode	Transient time (typ.)		Typical fast clamp current t1 … t2
±3 V, ±10 V, ±30 V, ±6	**t1**	**t2**	
±10 A	< 2 µs	< 20 µs	17 A
±3 A	< 2 µs	< 40 µs	5 A
±300 mA	< 2 µs	< 45 µs	600 mA
±30 mA	< 2 µs	< 80 µs	75 mA
±3 mA	< 2 µs	< 100 µs	20 mA
±300 µA	< 2 µs	< 100 µs	12 mA
±30 µA	< 2 µs	< 100 µs	< 10 mA

High-voltage mode	Transient time (typ.)		Typical fast clamp current t1 … t2
-60 V … +120 V	**t1**	**t2**	
±30 mA	< 2 µs	< 125 µs	75 mA
±3 mA	< 2 µs	< 250 µs	20 mA
±300 µA	< 2 µs	< 250 µs	12 mA
±30 µA	< 2 µs	< 250 µs	< 10 mA

FVI16 Kelvin measurement unit specifications

The Kelvin measurement unit is measuring the voltage drop of the force lines between HF# - HS# and LF# - LS#. A user defined test current (for example, 1 A, ~10 % of the maximum range) has to be forced through the DUT in order to measure the I * R voltage drop.



High force Kelvin drop voltage (Force drop voltage)

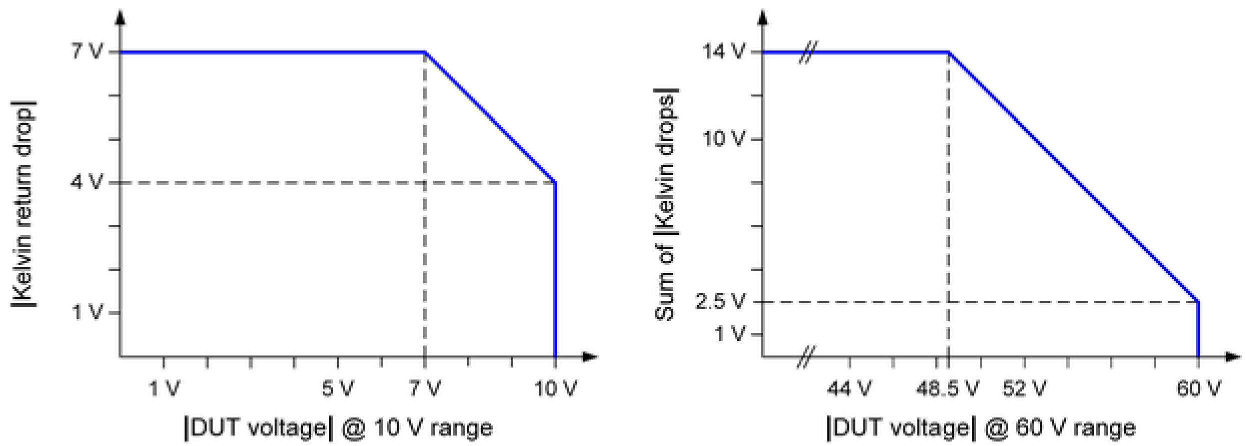
Voltage range	Condition	Programmable range	Accuracy	Example and comment
±3 V		±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g. @ 10 A = ±210 mV
±10 V		±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
±30 V		±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
±60 V	$\lvert V_{force} \rvert \leq 48.5$ V	±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
	$\lvert V_{force} \rvert > 48.5$ V	< 7 V see right diagram below		V_{force} # V_{clamp} for I_{force} operation
-60 V ... + 120		±1 V	±12 mV	

Low force Kelvin drop voltage (Return drop voltage)

Voltage range	Condition	Programmable range	Accuracy	Example and comment
±3 V		±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g. @ 10 A = ±210 mV

Voltage range	Condition	Programmable range	Accuracy	Example and comment
±10 V	Kelvin return threshold + $\lvert V_{force}\rvert \leq 14$ V	±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV V_{force} # V_{clamp} for I_{force} operation, a kelvin low flag is set in case of violation
	Kelvin return threshold + $\lvert V_{force}$ > 14 V	< 7 V see left diagram below		
±30 V		±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
±60 V	$\lvert V_{force}\rvert \leq 48.5$ V	±7 V	± (20 mV * I_{force}/A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV V_{force} # $_{clamp}$ for I_{force} operation
	$\lvert V_{force}\rvert > 48.5$ V	< 7 V see right diagram below		
-60 V ... + 120		N/A	N/A	

Maximum Kelvin drop voltage vs. DUT voltage



For the diagram above, you must also consider an internal voltage drop, which is defined by 1.0 Ω multiplied by the programmed force or clamp current.

Automatic disconnect for weak force to sense connection.

To protect probe card needles and test sockets, the VI source output gets automatically disconnected in case a weak force to sense connection leads to a Kelvin voltage drop exceeding the programmed threshold voltage and the VI is set to the 3 A or 10 A range[137].

Example:

Configuration

$R_{contact}$	= 2.0 Ω
Test current	= 1.0 A

[137] *SmarTest 7.5.2.1 and higher. With lower SmarTest versions, the channel is disconnected in all current ranges.*

Programmed Kelvin threshold limit	= 1.0 V

Voltage drop

$R_{contact}$ * Test current	= 2.0 Ω * 1.0 A = 2.0 V

Result

The voltage drop exceeds the programmed threshold of 1.0 V and the FVI16 channel gets disconnected.

FVI16 other specifications

Interface characteristics		
Compliance range if disconnected	HF, LF, HS, LS	±200 V
Input leakage current at disconnect state (typical)	HF, LF, HS, LS	< 20 nA
Lumped capacitance at disconnect state (typical)	LF to system GND	< 50 nF
	HS to LS	< 100 pF
	HS to LF	< 200 pF
	HF to LF	< 5 nF

Surge tracker	
Trigger	Voltage surges are detected against programmable minimum and maximum values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.
Minimum detectable pulse width	4 µs (typical)
Threshold voltage accuracy	See voltage digitizer specification (single shot)
Input and threshold range	Same as the measure voltage range (max. -60 … 120 V)

<table>
<tr><th colspan="2">DC profiling</th></tr>
<tr><td>DC profiling</td><td>Supported per channel for voltage and current profiles.
Profiling of the charge status of the on-board energy storages is supported.</td></tr>
<tr><td>Maximum sample rate</td><td>1 Msps
The maximum sample rate is available per channel group 1 … 8 and 9 … 16 and is shared by all channels of the group:
<table>
<tr><th>Channels</th><th>Maximum sample rate</th></tr>
<tr><td>1</td><td>1 Msps</td></tr>
<tr><td>2</td><td>500 ksps</td></tr>
<tr><td>3</td><td>333 ksps</td></tr>
<tr><td>...</td><td>...</td></tr>
</table>
Profiling the current of ganging groups is supported with a maximum sample rate of 125 ksps.</td></tr>
</table>

FVI16 pogo pin assignment

The E8026PVP pogo cable assembly connects the sixteen floating channels of the DC Scale FVI16 card to the DUT board.

Pogo pin assignment

Block A

	a	**b**	**c**	**d**
1	DNC	HF1	LF1	HS1
2	TMU_H1	LF1	HF1	LS1
3	TMU_L1	HF1	LF1	GUARD1
4	Separation			
5	DNC	HF2	LF2	HS2
6	TMU_H2	LF2	HF2	LS2
7	TMU_L2	HF2	LF2	GUARD2
8	Separation			
9	DNC	HF3	LF3	HS3
10	TMU_H3	LF3	HF3	LS3
11	TMU_L3	HF3	LF3	GUARD3
12	Separation			
13	DNC	HF4	LF4	HS4
14	TMU_H4	LF4	HF4	LS4
15	TMU_L4	HF4	LF4	GUARD4
16	Separation			
17	DNC	DNC	SAFETY1_GND	SAFETY1_5V

Orientation mark •

Block B

	a	**b**	**c**	**d**
1	DNC	HF5	LF5	HS5
2	TMU_H5	LF5	HF5	LS5
3	TMU_L5	HF5	LF5	GUARD5
4	Separation			
5	DNC	HF6	LF6	HS6
6	TMU_H6	LF6	HF6	LS6
7	TMU_L6	HF6	LF6	GUARD6
8	Separation			
9	DNC	HF7	LF7	HS7
10	TMU_H7	LF7	HF7	LS7
11	TMU_L7	HF7	LF7	GUARD7
12	Separation			
13	DNC	HF8	LF8	HS8
14	TMU_H8	LF8	HF8	LS8
15	TMU_L8	HF8	LF8	GUARD8
16	Separation			
17	DNC	DNC	DNC	DNC

Orientation mark •

Block C

	a	b	c	d
1	DNC	HF9	LF9	HS9
2	TMU_H9	LF9	HF9	LS9
3	TMU_L9	HF9	LF9	GUARD9
4	Separation			
5	DNC	HF10	LF10	HS10
6	TMU_H10	LF10	HF10	LS10
7	TMU_L10	HF10	LF10	GUARD10
8	Separation			
9	DNC	HF11	LF11	HS11
10	TMU_H11	LF11	HF11	LS11
11	TMU_L11	HF11	LF11	GUARD11
12	Separation			
13	DNC	HF12	LF12	HS12
14	TMU_H12	LF12	HF12	LS12
15	TMU_L12	HF12	LF12	GUARD12
16	Separation			
17	DNC	DNC	SAFETY2_GND	SAFETY2_5V

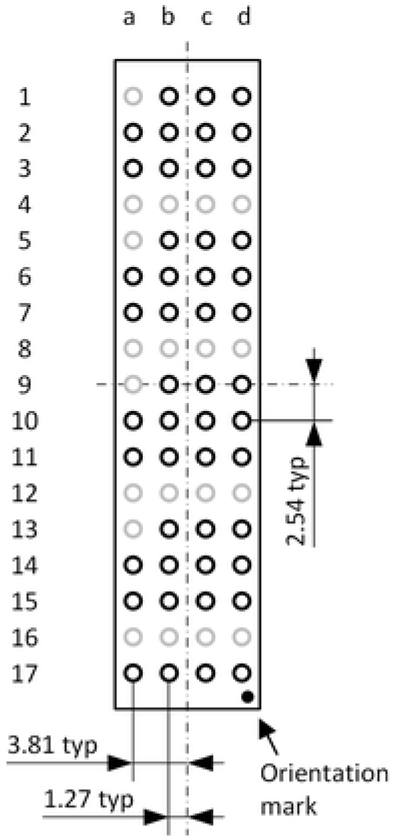
Orientation mark •

Block D

	a	b	c	d
1	DNC	HF13	LF13	HS13
2	TMU_H13	LF13	HF13	LS13
3	TMU_L13	HF13	LF13	GUARD13
4	Separation			
5	DNC	HF14	LF14	HS14
6	TMU_H14	LF14	HF14	LS14
7	TMU_L14	HF14	LF14	GUARD14
8	Separation			
9	DNC	HF15	LF15	HS15
10	TMU_H15	LF15	HF15	LS15
11	TMU_L15	HF15	LF15	GUARD15
12	Separation			
13	DNC	HF16	LF16	HS16
14	TMU_H16	LF16	HF16	LS16
15	TMU_L16	HF16	LF16	GUARD16
16	Separation			
17	DNC	DNC	DNC	DNC

Orientation mark •

Layout and dimension



See also *Pogo pad and via design*.

Pogo pin description

Pin	Description	DUT board connection
HF#	High force line of the channel	
HS#	High sense line of the channel	
LF#	Low force line of the channel	
LS#	Low sense line of the channel	
GUARD#	GUARD line of the HF# and HS# line	
TMU_H#	TMU high input	
TMU_L#	TMU low input	
SAFETY1_5V	Control line of the safety circuit for channel 1 … 8	Connect to an adequate +5.0 V supply via a safety interlock on the DUT board or the DUT interface.
SAFETY2_5V	Control line of the safety circuit for channel 9 … 16	
SAFETY1_GND	Return line of the safety circuit for channel 1 … 8	Connect to the return pin of the +5.0 V supply at SAFETY_P and to the DUT board ground.
SAFETY2_GND	Return line of the safety circuit for channel 9 … 16	

Pin	Description	DUT board connection
DNC	Do not connect pin	These pins are internally used and reserved. Do not connect these pins on the DUT board.
Separation	Separation row between the channels	To ensure that adjacent high voltage pins or traces have no effect on the DNC pins and Separation rows, adequate measures like sufficient insulation space have to be respected.

Notes on the pogo pin descriptions:

- Pin: # = Channel 1 … 16
- The DNC pins are internally connected and reserved. To ensure that adjacent high voltage pins or traces have no effect on the DNC pins, adequate measures like sufficient insulation space have to be respected.

⚠ CAUTION The force lines "HF#" and "LF#" are each wired to three pogo pins. These three pogo pins must be connected in parallel on the DUT board to ensure that the specifications for the card and the maximum ratings for the pogo pins are met.

DC Scale UHC4T Device Power Supply (E8024UH)

UHC4T is a device power supply card for the V93000 ATE platform.

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

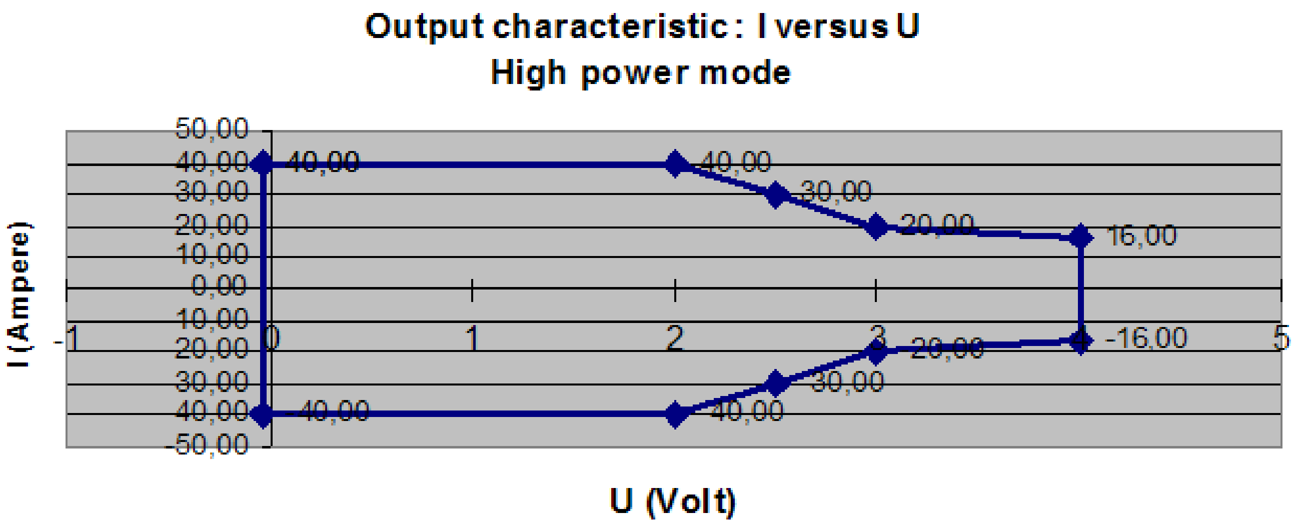
Configuration

Maximum number of channels per board	4
Maximum source/sink current per channel	40 A @ 2.0 V
Maximum source/sink current per channel	30 A @ 2.5 V
Maximum source/sink current per channel	20 A @ 3.0 V
Maximum source/sink current per channel	16 A @ 4.0 V
Maximum source/sink current per board	160 A @ 2.0 V
Maximum source/sink current per system	Maximum number of boards x 160 A
Maximum voltage	0 V to +4 V

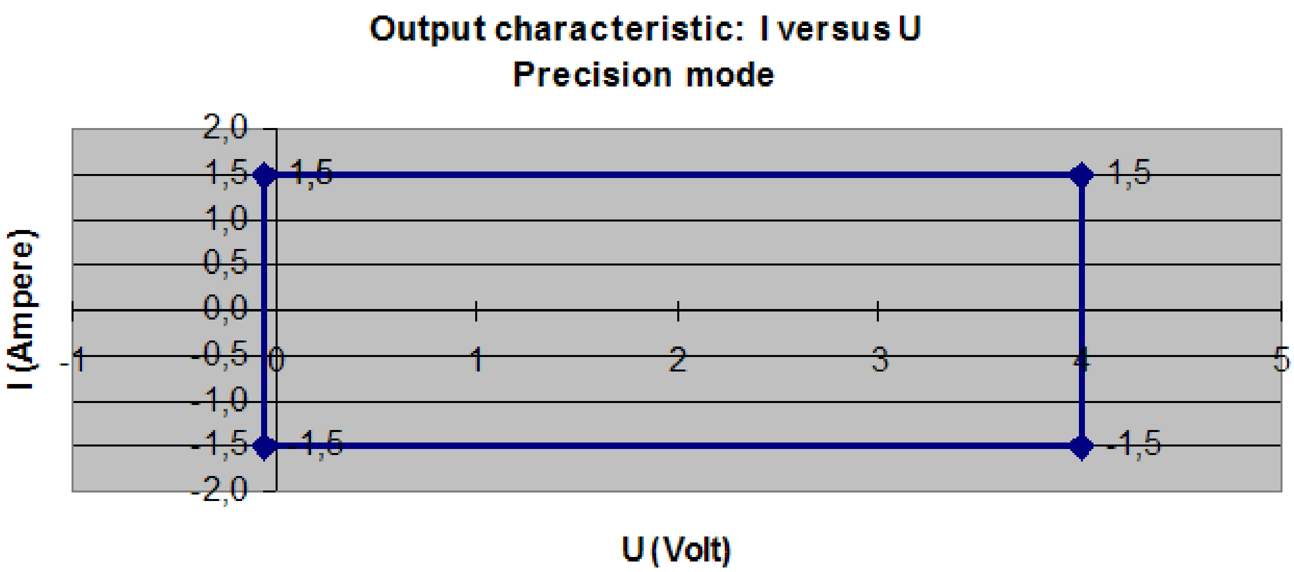
Channel and board maxima

	A test head[138]	**Compact test head[138]**	**Small test head[138]**	**Large test head**
Maximum number of boards per system	8	16	32	64
Maximum number of channels per system	32	64	128	256

[138] *Applies to test head with small DUT interface and DUT Scale interface*



UHC4T power diagram high power mode



UHC4T power diagram precision mode

- Parallel connection (ganging) possible for any number of successive channels within a card and any number of E8024UH cards within a test head.
- The maximum current per board cannot be exceeded.
- The UHC4T requires a universal DC/DC. An adjacent card must be a UHC4T again (located in a pair of uneven/even card cage slots).
- Static sink power on a board or both boards that are connected to the same DC/DC can be larger than the source power within these limits for each channel of a board:
  - ≤ 0.6 V : ≤ 40 A
  - ≤ 1.0 V: ≤ 7 A
  - ≤ 2.5 V: ≤ 4 A
  - ≤ 4.0 V: ≤ 2 A
- Dynamic sink power (dis-charge) is possible for supported capacitance.

- Either the high-power mode or the precision mode of a channel is active for force and measure. During the range transition from 1.5 to 10/40 A, both modes are active to avoid glitches.

  On the pogo block, four DUT_CAP_ON lines (DUT_CAP_ON_1 through DUT_CAP_ON_4) are available to switch the big blocking capacitance on the DUT board on or off.

  If the 40 A range is selected, the signal level is 0 V. If the 1.5 A range is selected, the signal level is 5 V.
- The UHC4T card requires no Enhanced Diagnostics Interconnect Board.

Supply voltage and current range specifications

	Range	Resolution	Accuracy
Voltage force	0 to 4.0 V	200 µV	±2.5 mV
Voltage measure	0 to 4.0 V	200 µV	±2 mV[139]
Current force (clamp)	10 to 40 A 0.5 to 10 A	2 mA 2 mA	±(100 mA + 0.5% of setting) ±(50 mA + 0.2% of setting)
	15 mA to 1.5 A	100 µA	±10 mA
Current force (n channels ganged)	n x 0.5 A to n x 40 A	n x 2 mA	±(n x 100 mA + 0.5% of total ganged current)
Current measurement	±40 A ±10 A	2 mA 2 mA	±(50 mA +0.5% of reading)[140] ±(20 mA +0.2% of reading)[140]
	±1.5 A	100 µA	±(1.5 mA +0.1 % of reading)[141]
	±50 mA	2.5 µA	±(50 µA +0.1 % of reading)[142]
	±2 mA	100 nA	±(2 µA +0.1 % of reading)[143]
Current measurement (n channels ganged)	n x 40 A	n x 2 mA	±(n x 50 mA +0.5% of total current)

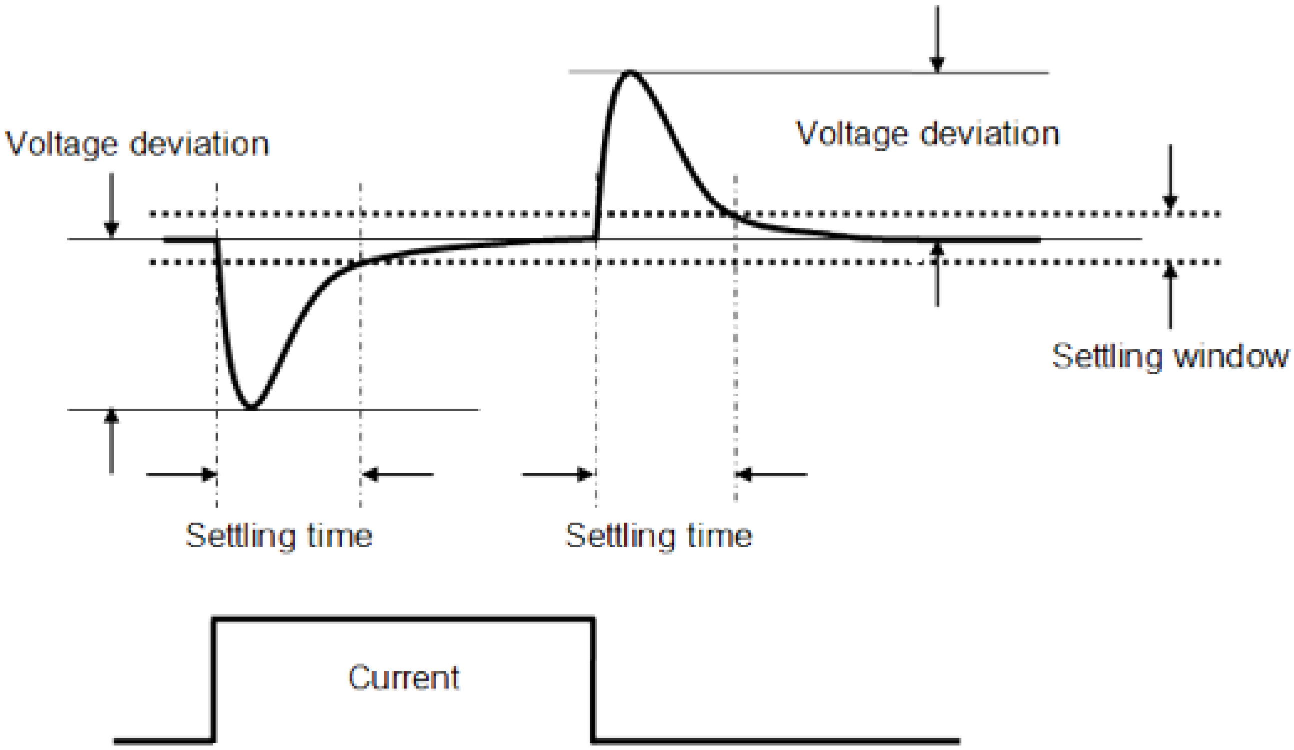
Fast unganging is supported, resulting in single-channel accuracy for Idd-measurements. This includes DUT capacitance switching using the four DUT_CAP_ON lines.

[139] *Requires 8 averaging samples under pattern control.*
[140] *Requires 64 averaging samples under pattern control.*
[141] *Requires 256 averaging samples under pattern control.*
[142] *Requires 256 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 100 µF.*
[143] *Requires 256 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 10 µF.*

Supported "Current force" and "Current measurement" range combinations

		Current force range, 10/40 A high-power mode	Current force range, 1.5 A precision mode
Current measurement range, high-power mode	10/40A	yes	no
Current measurement range, precision mode	1.5 A	no	yes
	50 mA	no	yes
	2 mA	no	yes

Explanation of terms



Explanation of terms

Settling time after load step (dI/dt = 2.5 A/μs)

At 2 V, settling into ±20 mV of programmed value, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	0 μs	0 μs
30 A to 10 A	0 μs	0 μs

At 2 V, settling into ±10 mV of programmed value, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	15 μs	15 μs
30 A to 10 A	18 μs	18 μs

Settling time after load step (dI/dt = 40 A/µs)

At 2 V, settling into ±20 mV of programmed value, single-channel operation:

Load step	1080 µF (bulk), 100 µF (ceramic)	2160 µF (bulk), 200 µF (ceramic)
10 A to 30 A	7 µs	7 µs
30 A to 10 A	7 µs	6 µs

At 2 V, settling into ±10 mV of programmed value, single-channel operation:

Load step	1080 µF (bulk), 100 µF (ceramic)	2160 µF (bulk), 200 µF (ceramic)
10 A to 30 A	9 µs	8 µs
30 A to 10 A	10 µs	7 µs

Settling time after load step (dI/dt = 400 A/µs)

At 2 V, settling into ±20 mV of programmed value, ganging four channels on one board:

Load step	4 mF (bulk), 200 µF (ceramic)	8 mF (bulk), 400 µF (ceramic)
40 A to 120 A	4 µs	5 µs
120 A to 40 A	9 µs	4 µs

At 2 V, settling into ±10 mV of programmed value, ganging four channels on one board:

Load step	4 mF (bulk), 400 µF (ceramic)	8 mF (bulk), 400 µF (ceramic)
40 A to 120 A	10 µs	7 µs
120 A to 40 A	13 µs	10 µs

Voltage deviation after load step (dI/dt = 2.5 A/µs)

At 2 V, single-channel operation:

Load step	270 µF (bulk)	540 µF (bulk)
10 A to 30 A	18 mV	18 mV
30 A to 10 A	18 mV	18 mV

Voltage deviation after load step (dI/dt = 40 A/µs)

At 2 V, single-channel operation:

Load step	1080 µF (bulk), 100 µF (ceramic)	2160 µF (bulk), 200 µF (ceramic)
10 A to 30 A	53 mV	35 mV
30 A to 10 A	55 mV	35 mV

Voltage deviation after load step (dI/dt = 400 A/µs)

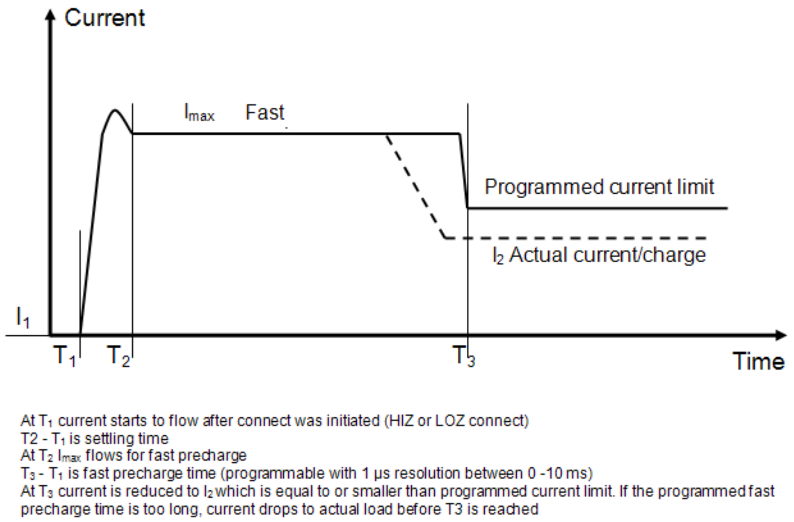
At 2 V, ganging four channels on one board:

Load step	4 mF (bulk), 200 µF (ceramic)	8 mF (bulk), 400 µF (ceramic)
40 A to 120 A	50 mV	35 mV
120 A to 40 A	80 mV	60 mV

Modes and warnings

Fast precharge	
Maximum supported current	40 A (independent of actual voltage setting) is forced for a programmable time (0-10 ms), independent of actual current limit settings, if the 10/40 A current force range is selected
Programmable timing resolution	1 µs
Precharge start	When DUT is connected
Mode selection	Explicitly enabled or disabled with a firmware command (default condition is disabled) or implicitly with a programmed time in an equation

- The fast precharge settings are identical for all channels of a gang group
- This mode is not enabled for voltage changes



Fast precharge mode

Peak mode	
Peak current	Peak current > 40 A for a few microseconds
Mode selection	Enabled, no control flag or alarm is set

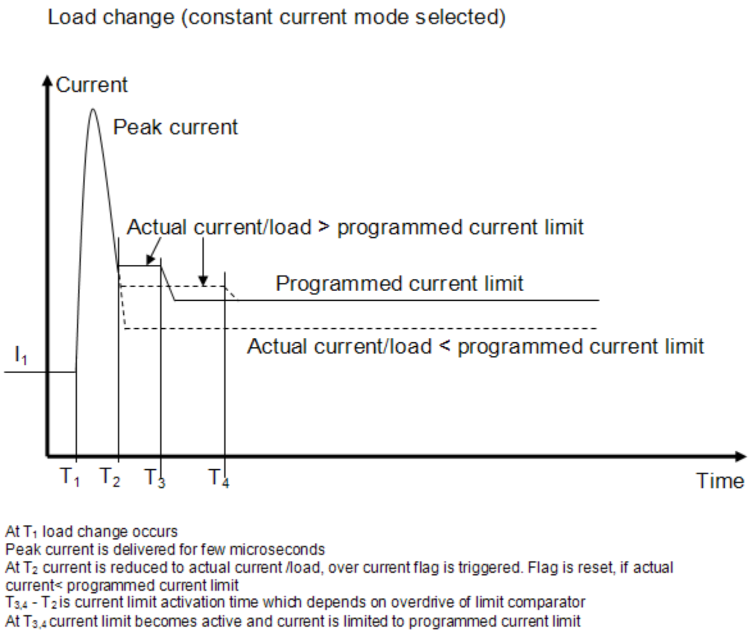
Current limit

Typical times to limit the current at a current limit of 10 A (one channel, no ganging, 1080 µF)

Load step	Response time
0 A to 40 A	600 µs

Load step	Response time
8 A to 24 A	1.3 ms
8 A to 16 A	3.2 ms

- Activation time of current limit depends on the actual load (overdrive of comparator)



Constant current mode

Constant current mode	
Constant current mode	Maximum allowed current is equal to actual programmed current limit

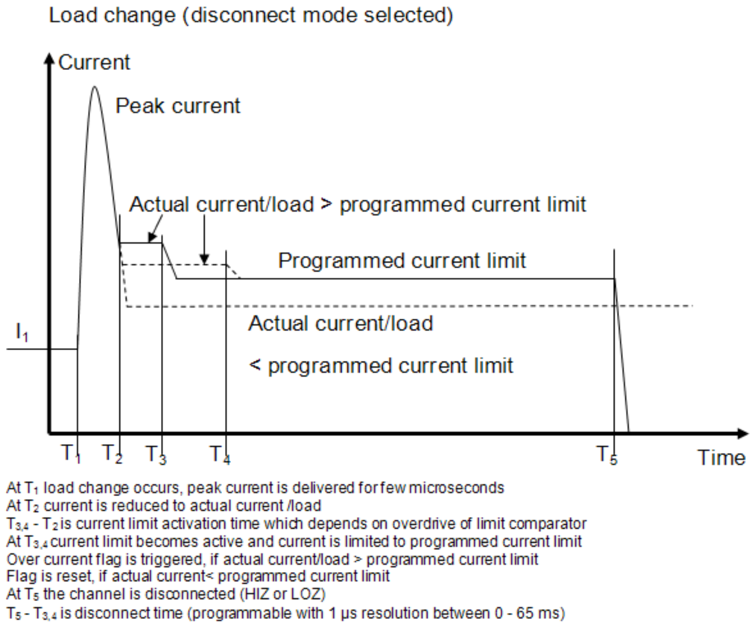
Disconnect mode	
Disconnect mode	If current exceeds the set current limit, the channel is disconnected
Disconnect delay time	Programmable (0 - 65 ms, timing resolution 1 µs)
Mode selection	Explicitly enabled or disabled with a firmware command (default condition is disabled) or implicitly with a programmed time in an equation

- Time delay counter is triggered by the overcurrent flag.
- The settings are identical for all channels of a gang group.
- This mode is selectable in all current force ranges.
- The safety line is not activated (same behavior as HC-DPS).

Voltage drop protection	
Allowed voltage drop	Individually programmable for force and return path between 0 - 500 mV, programming resolution is 1 mV
Total allowed voltage drop	500 mV

Voltage drop protection	
Mode selection	Always on, in all current ranges, except for the time the fast precharge mode is on
Default voltage limit	250 mV for both the force and return path

- Force voltage at output of UHC4T is continuously compared to voltage at charge. If the difference is above the programmed limit, the output channel is disconnected. The voltage drop can be measured with the UHC4T.
- There are separate comparators for both the force and the return line on both the source and the sink path (four in total).
- The safety line is not activated in case of alarm (same behavior as HC-DPS).



Disconnect mode

Other protection mechanisms (one line per board)	
Safety line	Connected to DUT board via pogo block. Internally pulled high to +5V from DUT board with a resistor of 46.4k. When pulled low, all channels of UHC are disconnected. Can be used to simultaneously disconnect all UHC4T channels by an external alarm switch.
+5 V from DUT board	If not applied, all UHC4T channels are disconnected. Can be used to detect improperly connected DUT board.

Other performance characteristics

Disconnect state	
Input leakage current	< 100 nA @ ≤ 3.4 V (AC relay open)

Voltmeter mode (high impedance mode)	
Voltage range	0 V to +4 V

Voltmeter mode (high impedance mode)	
Voltage resolution	200 µV
Voltage accuracy	±2 mV
Input leakage current	< 100 nA @ ≤ 3.4V (device connected to force and sense lines)

Differential mode (using force sense and ground sense)	
Voltage range	Same as voltage measure
Voltage resolution	Same as voltage measure
Voltage accuracy	Same as voltage measure

Trigger functions	
Trigger	Sequencer#controlled (no external trigger pin required)

Data analysis	
Data analysis	Raw data is stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board.
Result analysis	Built-in result analysis available that indicates the measurement data is pass or fail against a user specified test limit.

Compliance range (at disconnect state)	
Maximum external voltage at force input	-3 V to +10 V
Maximum external voltage at sense input	-3 V to +10 V
Maximum external voltage at sampling input	-3 V to +10 V
Maximum external voltage at safety line	-3 V to +5.5 V
Maximum external voltage at +5 V DUT signal	-3 V to +5.5 V

Settling time after re-programming (settling within given band of programmed step or within accuracy band, 2 x 270 µF capacitive charge)	
Voltage force (0-2 V step)	200 µs (± 1% band)
Current force / measure (0-10 A step)	40 µs (+0 -20% band) 4 ms (± 1% band)

Charge capacitance range	
Minimum required external capacitance	270 µF per channel for high power mode 1 µF per channel for precision mode

Charge capacitance range	
Maximum supported external capacitance with stability maintained	10 mF per channel
Recommended external capacitance	1 - 2 mF, see also the explanation for the voltage/ current specifications, above

Blocking caps (examples, both types are required on DUT board)

Tantalum (example for bulk capacitance)	
Kemet	T530D477M2R5A(1)E010 - 470 μF/2.5V
Panasonic	OSCON 4SVPC560MX - 560 μF/4V

Ceramic (example)	
TDK	C1608X5R1E105K (1uF/25V/0603/X5R-Ceramic), array of 25 each
TDK	C1608X7R1H104K (100nF/50V/0603/X7R-Ceramic), array of 25 each

Power ramp at probe	
Probe card protection	User-programmable sequence for both shorts and open-check, using the precision mode

Thermal control	
Current monitor	Current monitor output per channel provides voltage proportional to the output current, 1 V per 10 A current, reference potential is GSn- line (Ground Sense). **Note** Sense-lines are sensitive to noise. It it is recommended to use the GND-plane close to the connection of the GSn-line.
Typical accuracy	±5 mV ± 1% of reading

DC Scale XPS256 / XPS128 / XPS128+HV Universal DPS/VI Card

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Hardware support in SmarTest

XPS256 variant		SmarTest 7	SmarTest 8
XPS256	(EK751HC)	7.10.1 and higher	8.3.4 and higher
XPS128	(EK751HCH)	not supported	8.5.3 and higher
XPS128+HV	(EK751HCV)	not supported	8.5.4 and higher

XPS256 overview

General specifications		
Maximum channel count		
	XPS256	256 channels per card
	XPS128	128 channels per card
	XPS128+HV	128 channels per card
Maximum source current per channel		
	XPS256 / XPS128	1.0 A @ -2.5 V … +3.0 V 0.6 A @ -2.5 V … +7.0 V
	XPS128+HV	1.0 A @ -2.5 V … +3.0 V 0.6 A @ -2.5 V … +7.0 V 0.25 A @ -10.0 V … +15.0 V 0.15 A @ -1.0 V … +24.0 V
Maximum sink current per channel		
	XPS256 / XPS128	1.0 A @ -2.0 V … +3.0 V 0.6 A @ -2.5 V … +7.0 V
	XPS128+HV	1.0 A @ -2.0 V … +3.0 V 0.6 A @ -2.5 V … +7.0 V 0.25 A @ -10.0 V … +15.0 V 0.15 A @ -1.0 V … +24.0 V
Maximum source current per card		
	XPS256	256 A @ -2.5 V … +2.0 V [144]
	XPS128 / XPS128+HV	128 A @ -2.5 V … +2.0 V [144]
Maximum sink current per card		
	XPS256	230 A @ -2.1 V … +4.0 V [144]
	XPS128 / XPS128+HV	115 A @ -2.1 V … +4.0 V [144]
Voltage range		
	XPS256 / XPS128	-2.5 V … +7.0 V
	XPS128+HV	-10.0 V … +24.0 V

Calibration, operation, environmental	
Warm-up time	60 minutes
Basic maintenance period	6 months

[144] *Combined voltage drop on force and GND line not exceeding ± 0.2 V*

Calibration, operation, environmental	
Base calibration period (traceable calibration)	6 months
Calibration period (system auto adjustment) [145]	3 months
Operating temperature range	15 °C ... 30 °C (59 °F ... 86 °F)
Temperature range for guaranteed specifications	20 °C ... 30 °C (68 °F ... 86 °F)
Maximum humidity at 30 °C	< 70 % R. H., non-condensing
Storage temperature range (without water)	-40 °C ... +70 °C (-40 °F ... 158 °F)

System configuration		**CX**	**SX**	**LX**
Maximum number of cards per system		9	18	27
Maximum number of channels per system	XPS256	2304 [147]	4608 [148]	6912 [148]
	XPS128 / XPS128+HV	1152 [146]	2304 [146]	3456 [146]

Other specifications
• Maximum source/sink current per card cannot be exceeded. • Requires E7010F, Rev. 11 or higher or EK855AD Enhanced Diagnostics Interconnect Board.

[145] *Valid at ambient temperature within ±2.5 K of calibration temperature.*
[146] *With EK751VHDB pogo cable assembly and DUT Scale interface*
[147] *Max. 2048 channels with DUT Scale interface and EK751HDA pogo cable assembly*
[148] *Max. 4096 channels with DUT Scale interface and EK751VHDA pogo cable assembly (max. 2048 channels with EK751HDA)*

XPS256 voltage specifications

Voltage force	Range	Resolution	Accuracy [149] [152]
XPS256 / XPS128	-0.5 V ... +1.5 V	10 µV	± (150 µV + 0.02 % of setting)
	-2.5 V ... +7.0 V	50 µV	± (400 µV + 0.02 % of setting)
XPS128+HV	-0.5 V ... +1.5 V	10 µV	± (150 µV + 0.02 % of setting)
	-2.5 V ... +7.0 V	50 µV	± (400 µV + 0.02 % of setting)
	-10.0 V ... +15.0 V	150 µV	± (1 mV + 0.02 % of setting)
	-1.0 V ... +24.0 V	150 µV	± (1 mV + 0.02 % of setting)

Voltage measurement	Range	Resolution	Accuracy [149] [151]
XPS256 / XPS128	-0.5 V ... +1.5 V	10 µV	± (150 µV + 0.02 % of value)
	-2.5 V ... +7.0 V	50 µV	± (400 µV + 0.02 % of value)
XPS128+HV	-0.5 V ... +1.5 V	10 µV	± (150 µV + 0.02 % of value)
	-2.5 V ... +7.0 V	50 µV	± (400 µV + 0.02 % of value)
	-10.0 V ... +15.0 V	150 µV	± (1 mV + 0.02% of value)
	-1.0 V ... +24.0 V	150 µV	± (1 mV + 0.02% of value)

Voltage clamp	Range	Resolution	Accuracy [150]
XPS256 / XPS128	-0.5 V ... +1.5 V	10 µV	± (500 µV + 0.1 % of setting)
	-2.5 V ... +7.0 V	50 µV	± (500 µV + 0.1 % of setting)
XPS128+HV	-0.5 V ... +1.5 V	10 µV	± (500 µV + 0.1 % of setting)
	-2.5 V ... +7.0 V	50 µV	± (500 µV + 0.1 % of setting)
	-10.0 V ... +15.0 V	150 µV	± (3 mV + 0.1 % of setting)
	-1.0 V ... +24.0 V	150 µV	± (3 mV + 0.1 % of setting)

[149] Measurement conditions VBw = 30 kHz, IBw = 10 kHz, needs open loop protection = "none", needs SmarTest 8.5.1 / SmarTest 7.10.3 or later.
[150] Measurement conditions VBw = 30 kHz, IBw = 10 kHz.
[151] Requires 40 samples (20 µs).
[152] Max cap load < 1 mF / channel

XPS256 current specifications

Current force	Range	Resolution	Accuracy [154] [157]
	± 1.0 A	10 µA	± (1 mA + 0.15 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
	± 100 mA	1 µA	± (100 µA + 0.15 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
	± 10 mA	100 nA	± (10 µA + 0.15 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
	± 1 mA	10 nA	± (1 µA + 0.15 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
	± 100 µA	1 nA	± (250 nA + 0.15 % of setting + 1 nA/V * $\lvert V_{act} \rvert$)
	± 10 µA	100 pA	± (50 nA + 0.15% of setting + 1 nA/V * $\lvert V_{act} \rvert$)
Current force (n channels gange	± n * 1.0 A	n * 10 µA	± (n * 1 mA + 0.15 % of setting + n * 1 nA/V * $\lvert V_{act} \rvert$)

Current measurement	Range	Resolution	Accuracy [155] [158] [157]
	± 1.0 A	10 µA	± (1 mA + 0.1 % of value + 1 nA/V * $\lvert V_{act} \rvert$)
	± 100 mA	1 µA	± (100 µA + 0.1 % of value + 1 nA/V * $\lvert V_{act} \rvert$)
	± 10 mA	100 nA	± (10 µA + 0.1 % of value + 1 nA/V * $\lvert V_{act} \rvert$)
	± 1 mA	10 nA	± (1 µA + 0.1 % of value + 1 nA/V * $\lvert V_{act} \rvert$)
	± 100 µA	1 nA	± (250 nA + 0.1 % of value + 1 nA/V * $\lvert V_{act} \rvert$) [159]
	± 10 µA	100 pA	± (50 nA + 0.1 % of value + 1 nA/V * $\lvert V_{act} \rvert$) [159]
Current measure (n channels gange	± n * 1.0 A	n * 10 µA	± (n * 1 mA + 0.1 % of value + n * 1 nA/V * $\lvert V_{act} \rvert$))

Current clamp	Range	Resolution	Accuracy [156]
	10 mA … 1.0 /	10 µA	± (2 mA + 0.2 % of setting)
	1 mA … 100 n	1 µA	± (200 µA + 0.2 % of setting)
	100 µA … 10 ı	100 nA	± (20 µA + 0.2 % of setting)
	10 µA … 1 mA	10 nA	± (2 µA + 0.2 % of setting)
	1 µA … 100 µ/	1 nA	± (400 nA + 0.2 % of setting)
	100 nA … 10 µ	100 pA	± (70 nA + 0.2 % of setting)
Current clamp (n channels gange	n * (10 mA …	n * 10 µA	± (n * 2 mA + 0.2 % of setting)
Clamp response time (typical)	< 35 µs [153]		

[153] *At Ibandwidth = 50 kHz, into programmed value ± 10 % of Irange, loadstep mode standard*

Hardware current limiter	
Short circuit current	< 1.8 A

[154] *Measurement conditions VBw = 30 kHz, IBw = 10 kHz, needs open loop protection = "none", needs SmarTest 8.5.1 / SmarTest 7.10.3 or later.*

[155] *Measurement conditions VBw = 30 kHz, IBw = 10 kHz, measure bandwidth = 30 kHz, needs open loop protection = "none", needs SmarTest 8.5.1 / SmarTest 7.10.3 or later.*

[156] *Measurement conditions VBw = 30 kHz, IBw = 10 kHz*

[157] *V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).*

[158] *Requires 160 samples (80 µs).*

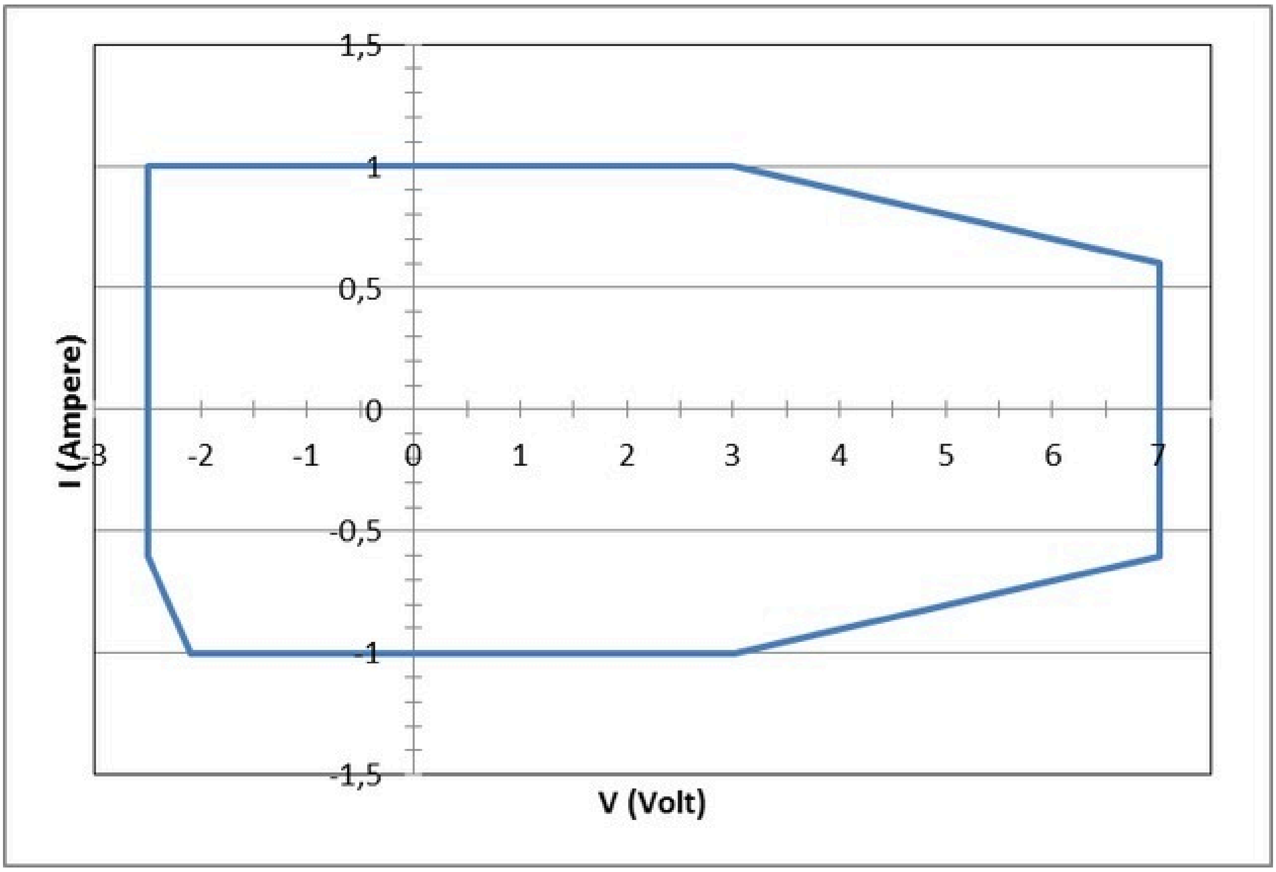
[159] *Requires 640 samples (320 µs).*

XPS256 voltage and current capability specifications

Continuous voltage and current capabilities per channel

XPS256 / XPS128 / XPS128+HV

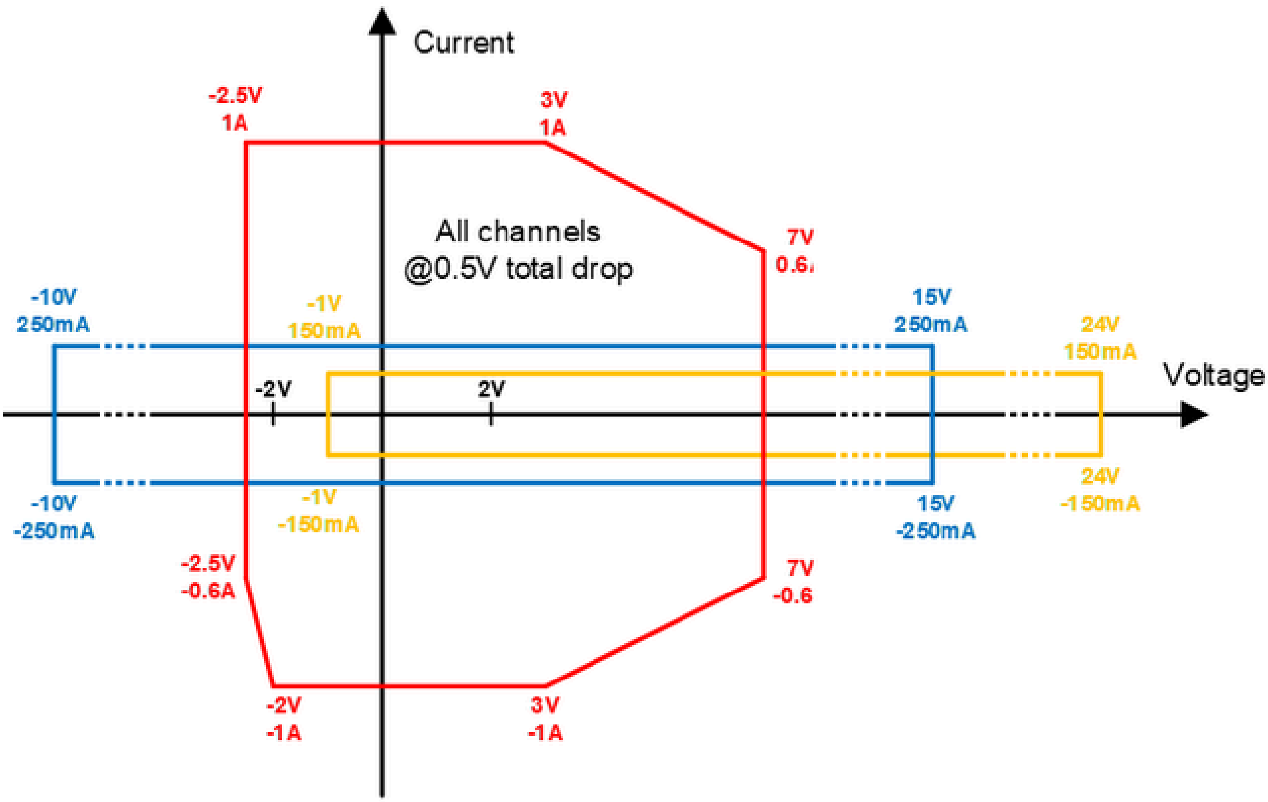
The following diagram shows the capability for a single channel in the 1.0 A high-current range for voltages up to 7 V.

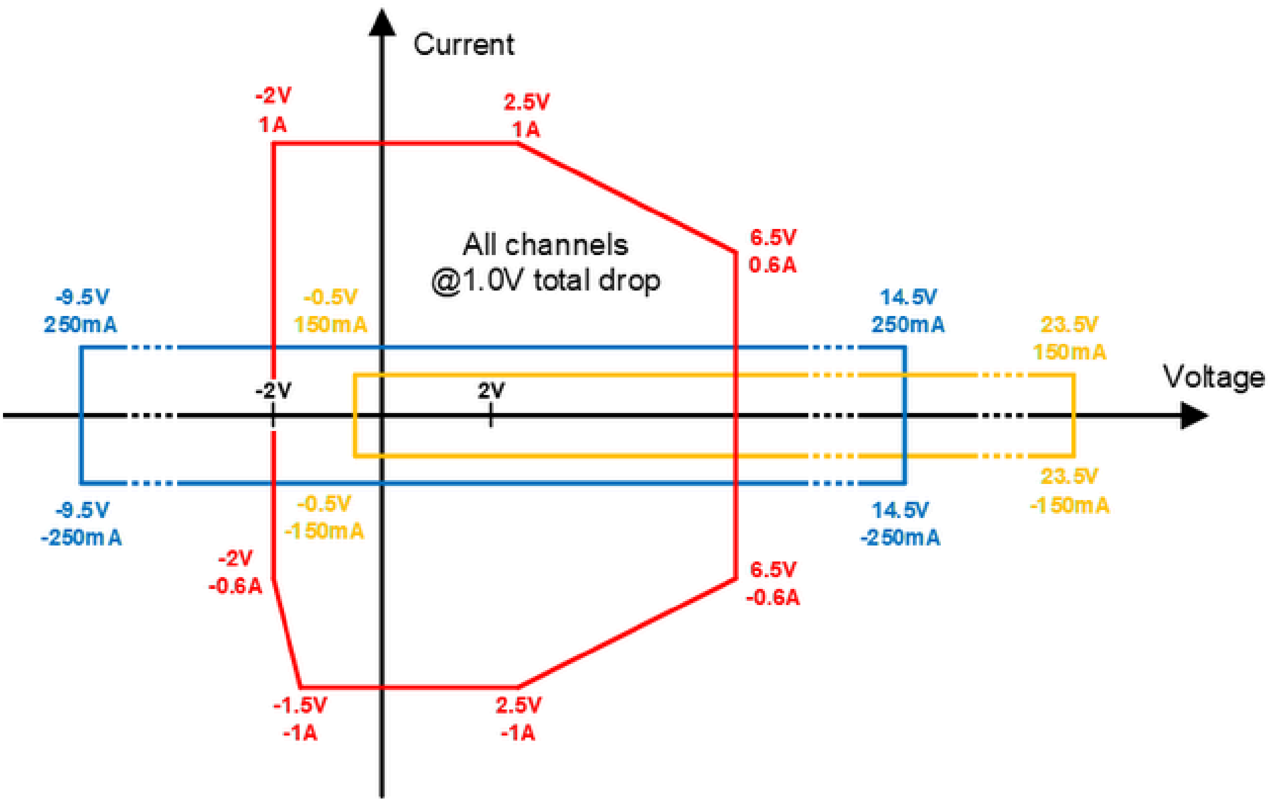


The power diagram applies to an additional voltage drop in the force and return path beyond the pogo pin of total ≤ 0.5 V.

XPS128+HV

The following diagrams apply to the 1.0 A high-current and high-voltage ranges of XPS128+HV.





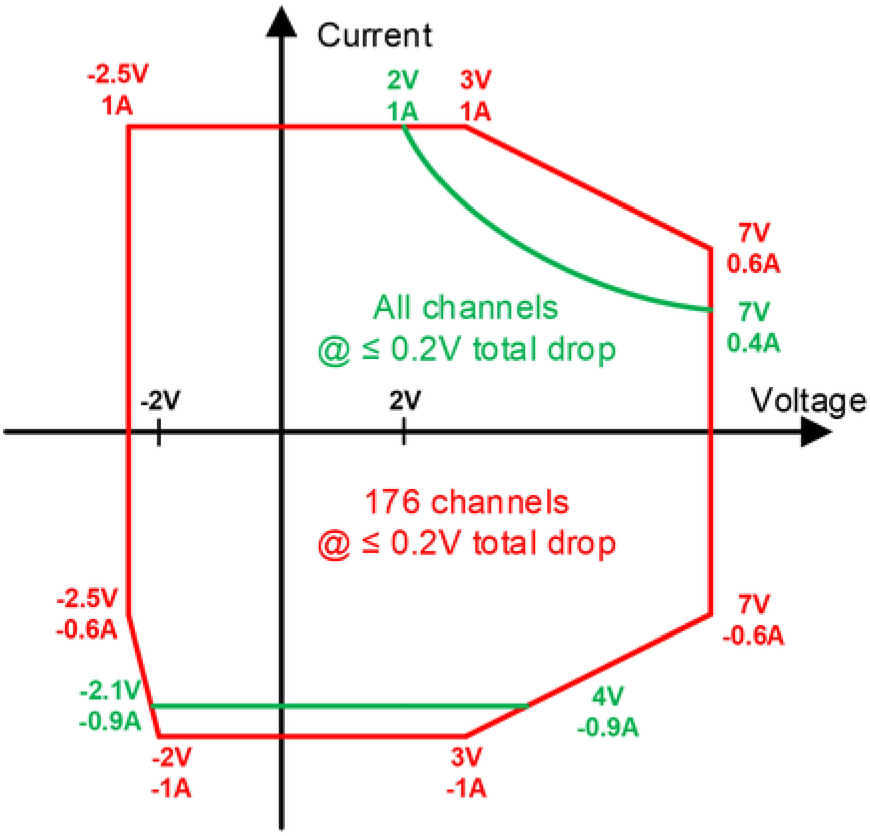
The upper diagram applies to an additional voltage drop in the force and return path beyond the pogo pin of total ≤ 0.5 V while the lower diagram applies to an additional voltage of total ≤ 1.0 V.

Maximum continuous voltage and current capability per card

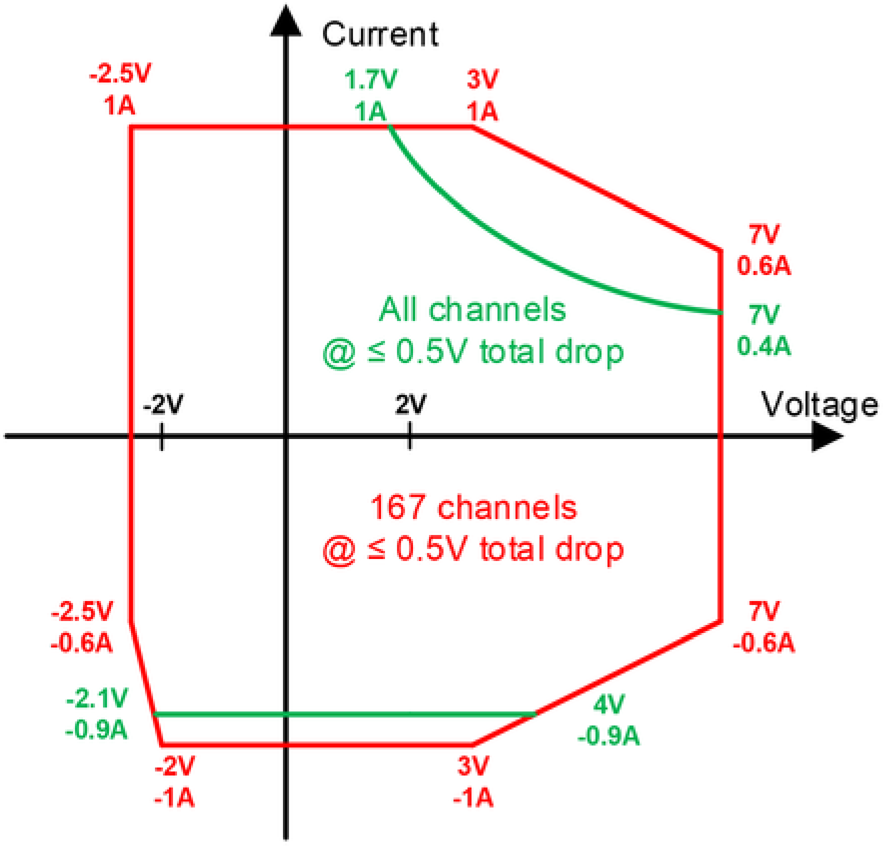
The following diagrams show the capability per card for a different number of active channels and different maximum total voltage drop on the force lines P and GND from the pogo pins to the DUT.

Note that for XPS128 and XPS128+HV, the channel count limitation does not apply. The two variants support the maximum continuous voltage and current capability for all 128 channels as defined by the red lines.

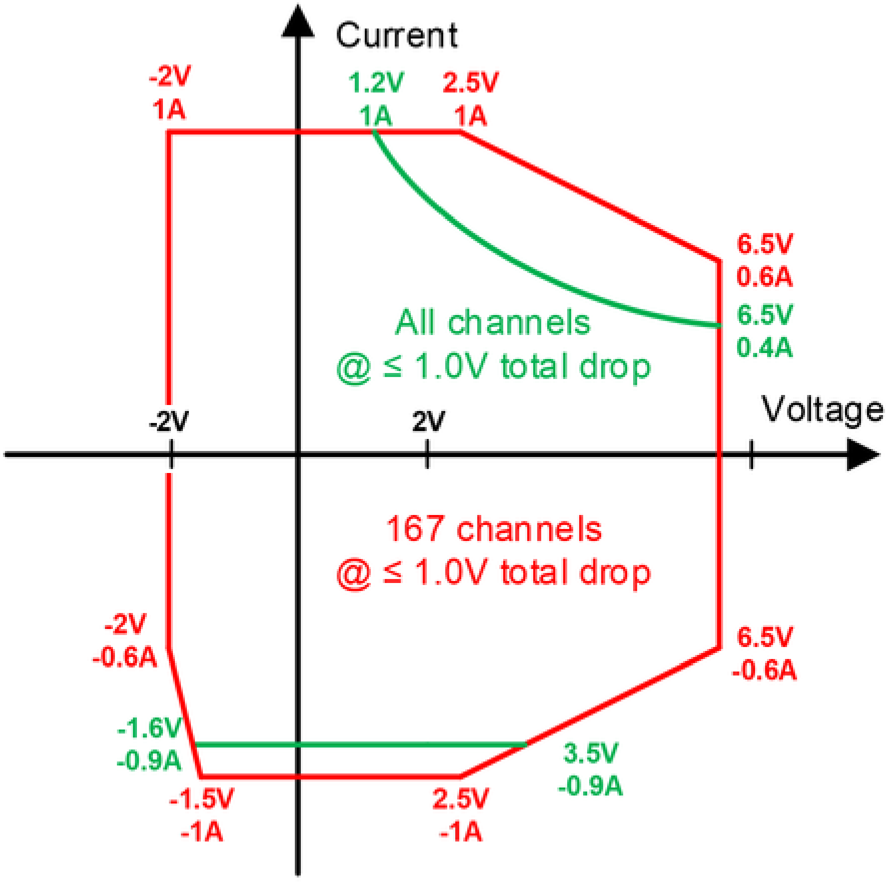
All channels / 176 channels @ 0.2 V maximum total voltage drop for force and return path



All channels / 167 channels @ 0.5 V maximum total voltage drop for force and return path

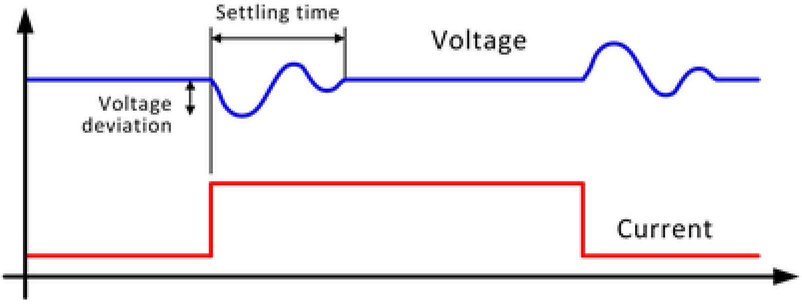


All channels / 167 channels @ 1.0 V maximum total voltage drop for force and return path


Current
-2V
1A
1.2V
1A
2.5V
1A
6.5V
0.6A
6.5V
0.4A
All channels
@ ≤ 1.0V total drop
-2V
2V
Voltage
167 channels
@ ≤ 1.0V total drop
-2V
-0.6A
6.5V
-0.6A
-1.6V
-0.9A
3.5V
-0.9A
-1.5V
-1A
2.5V
-1A

XPS256 load step response specifications

Explanation of terms:



Load step response [160]	Regulation mode	Load step	100 µF [164]	47 µF [164]	10 µF [161]
Settling time [162]	Fast	50 % [165]	20 µs	15 µs	10 µs
		80 % [166]	20 µs	15 µs	10 µs
	Standard	50 % [165]	50 µs	30 µs	20 µs
		80 % [166]	55 µs	35 µs	25 µs
Maximum voltage deviation [163]	Fast	50 % [165]	30 mV	40 mV	85 mV
		80 % [166]	55 mV	70 mV	120 mV
	Standard	50 % [165]	60 mV	65 mV	110 mV
		80 % [166]	90 mV	110 mV	140 mV

[160] *Typical at pogo pin, in n * ±1.0 A range, clamp setting n * 1 A (n = number of channels ganged), only valid for -0.5 V … +1.5 V, -2.5 V … +7.0 V ranges*
[161] *Voltage bandwidth: 30 kHz*
[162] *At 1 V, settling into ± 10 mV of programmed value, any number of channels ganged*
[163] *At 1 V, any number of channels ganged*
[164] *Voltage bandwidth: 50 kHz*
[165] *Verified at 25 % to 75 % load step*
[166] *Verified at 10 % to 90 % load step*

XPS256 AWG, digitizer, and comparator specifications

Source waveform mode (AWG)	
Regulated mode:	
Max. step rate	2 Msps (million samples per second) [170]
Analog bandwidth (-3 dB, 1 V swing)	30 kHz (typ.)
Maximum number of samples	14 Msamples (million samples) [167] [168]
Accuracy	see Voltage force / Current force specifications

Measure waveform mode (Digitizer)	
Voltage sampling:	
Sample rate	2 Msps (million samples per second) [170]
Signal path bandwidth (-3 dB)	500 kHz (typ.)
Maximum number of samples	22 Msamples (million samples) [172]
Accuracy	see voltage measure specifications [169]
Current sampling:	
Sample rate	2 Msps (million samples per second) [170]
Signal path bandwidth (-3 dB)	500 kHz (typ.)
Maximum number of samples	22 Msamples (million samples) [172]
Accuracy	see current measure specifications [169]

Voltage compare mode [171]	
Minimum detectable pulse width	1 µs (typ.)
Threshold voltage accuracy	Same as voltage measure (1 kHz filter)
Input and threshold range	Same as voltage measure

Current compare mode [171]	
Minimum detectable pulse width	1 µs (typ.)
Threshold current accuracy	Same as current measure (1 kHz filter)
Input and threshold range	Same as current measure

[167] *Requires SmarTest 8, 1 Msamples max. with SmarTest 7*
[168] *64 bits per sample (256 bits for SmarTest 7) in total channel memory (896 Mbit per channel), channel memory will be shared with other DC actions and ganged channels within a group of 8 channels*
[169] *With 20 kHz digital filter*

[170] *Requires SmarTest 8, 1.5 Msps max. with SmarTest 7*
[171] *Requires SmarTest 8*
[172] *40 bits per sample (max. 11 Msamples, 80 bits per sample for SmarTest 7) in total channel memory (896 Mbit per channel), channel memory will be shared with other DC actions (max 16M per action)*

XPS256 other performance characteristics

Load capacitance range	
Minimum required external capacitance with stability maintained	0 nF [173]
Maximum supported external capacitance with stability maintained	1000 µF per channel [173] (n * 1000 µF if ganged)

Output noise	
Output noise (spurious up to 20 MHz, typical) [174]	100 µV rms [175] 600 µV rms [176]

Slew rate control (Voltage)

On-, off-, change level-rate individually controlled for each channel

	1.5 V range	**7.0 V range**	**15.0 V range**	**24.0 V range**
Minimum programmable slew rate	10 µV / µs	45 µV / µs	120 µV / µs	120 µV / µs
Maximum programmable slew rate	2 V / µs	2 V / µs	2 V / µs	2 V / µs

Slew rate control (Current)

Change level-rate individually controlled for each channel [177]

Range	**Min**	**Max**
± 1.0 A	10.0 µA/µs	200.0 mA/µs
± 100 mA	1.0 µA/µs	20.0 mA/µs
± 10 mA	100.0 nA/µs	2.0 mA/µs
± 1 mA	10.0 nA/µs	200.0 µA/µs
± 100 µA	1.0 nA/µs	20.0 µA/µs
± 10 µA	100.0 pA/µs	2.0 µA/µs

Disconnect mode	
Disconnect mode	If current reaches set current limit, the channel is disconnected

[173] *In standard regulation mode, Vbandwith = 10 kHz*
[174] *For single channel, in 1A range*
[175] *At 100 µF per channel capacitive load*
[176] *At 0 nF capacitive load*
[177] *Min/max values valid for single channel, for n x ganging, numbers must be multiplied by n*

Disconnect mode		
Disconnect delay time [178]	Range	Resolution
	0 µs … 127 µs	1 µs
	> 127 µs … 1.27 ms	10 µs
	> 1.27 ms … 12.7 ms	100 µs
	> 12.7 ms … 127 ms	1 ms
	> 127 ms … 1.27 s	10 ms

Disconnect state	
Max. input leakage current (AC relay open, typical)	± (10 nA + 1nA / V) force and sense connected
	± (1 nA + 1nA / V) only sense connected
Voltmeter mode (high impedance mode)	
Voltage range	Same as voltage measure
Voltage resolution	Same as voltage measure
Voltage accuracy	Same as voltage measure
Max. input leakage current (typical)	± (1 nA + 1nA / V)
Max. input voltage (sense / ground sense)	-0.5 V … +1.5 V in 1.5 V range
	-2.5 V … +7.0 V in 7.0 V range
	-10.0 V … +15.0 V in 15.0 V range
	-1.0 V … +24.0 V in 24.0 V range
Max. differential voltage (sense to ground sense)	-0.5 V … +1.5 V in 1.5 V range
	-2.5 V … +7.0 V in 7.0 V range
	-10.0 V … +15.0 V in 15.0 V range
	-1.0 V … +24.0 V in 24.0 V range

Compliance range (at disconnect state)	
Min/max external voltage at force input	-4.5 V … +7.5 V in 1.5 V range
	-4.5 V … +7.5 V in 7.0 V range
	-13.5 V … +18.5 V in 15.0 V range
	-4.5 V … +27.5 V in 24.0 V range
Min/max external voltage at sense input	-4.5 V … +7.5 V in 1.5 V range
	-4.5 V … +7.5 V in 7.0 V range
	-13.5 V … +18.5 V in 15.0 V range
	-4.5 V … +27.5 V in 24.0 V range

[178] Time delay counter is triggered by the overcurrent flag. The settings are identical for all channels of a gang group.

Overvoltage protection	
Maximum voltage range	-80 V … +80 V
Maximum energy of voltage spike	20 mJ
Maximum current	1.0 A

Voltage drop protection / measurement	
Allowed voltage drop	Individually programmable for force and return path between 0 and 500 mV, programming resolution is 1 mV Two level sets can be programmed
Total allowed voltage drop	1000 mV
Voltage drop measurement accuracy (relative)	± 4 mV [179]
Voltage drop measurement accuracy (absolute)	± (20 mV + 50 mV / A)
Voltage drop measurement resolution	900 µV

DC profiling	
DC profiling	Supported per channel for simultaneous voltage, current, and voltage drop profiling [180]
Maximum sample rate	2 Msps (million samples per second) [181]
Maximum number of samples	176 Msamples (million samples) [182]
Input signal bandwidth	500 kHz (typ.)

Surge tracker	
Trigger	Voltage or current surges are detected against programmable min. and max. values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.
Input signal bandwidth	500 kHz (typ.)
Threshold voltage accuracy (typical)	± 10 mV in 1.5 V range ± 10 mV in 7.0 V range ± 30 mV in 15.0 V range ± 30 mV in 24.0 V range
Threshold current accuracy (typical)	± 10 % of selected current range

[179] *Delta measurements on same channel at same operating conditions (i.e. force current, same auto-calibration). Requires 40 averages.*

[180] *Simultaneous profiling requires SmarTest 7.10.4 or higher or SmarTest 8.4.2 or higher.*

[181] *Requires SmarTest 7.10.5 / 8.4.2 or higher, valid for profiling one parameter only*

[182] *When profiling on one channel within any group of 8 channels. 22 Msamples when profiling on all 8 channels within a group of 8 channels.*

XPS256 pogo block specifications

EK751HDA - High Density Pogo Cable for EK751 (16 x 16 channels)

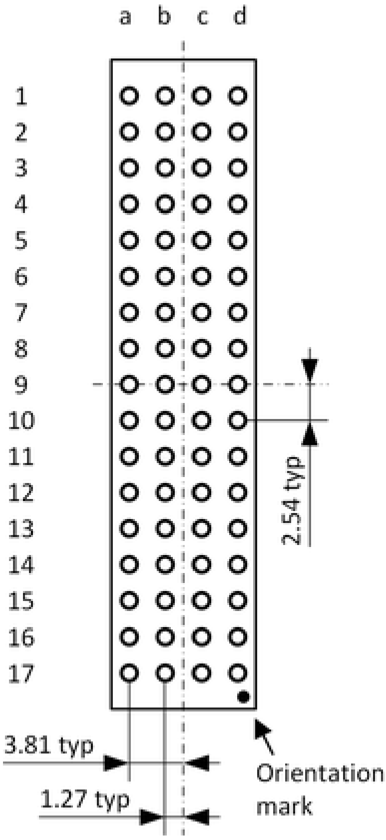
High-density (HD) pogo block

16 channels per pogo block

	a	**b**	**c**	**d**
1	GS2	P1	GND	PS1
2	PS2	GND	P2	GS1
3	GS4	P3	GND	PS3
4	PS4	GND	P4	GS3
5	GS6	P5	GND	PS5
6	PS6	GND	P6	GS5
7	GS8	P7	GND	PS7
8	PS8	GND	P8	GS7
9	GS10	P9	GND	PS9
10	PS10	GND	P10	GS9
11	GS12	P11	GND	PS11
12	PS12	GND	P12	GS11
13	GS14	P13	GND	PS13
14	PS14	GND	P14	GS13
15	GS16	P15	GND	PS15
16	PS16	GND	P16	GS15
17	GND	NC	GND	NC

Orientation mark •

Layout and dimension



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EK751VHDA - Very High Density Pogo Cable for EK751 (8 x 32 channels)

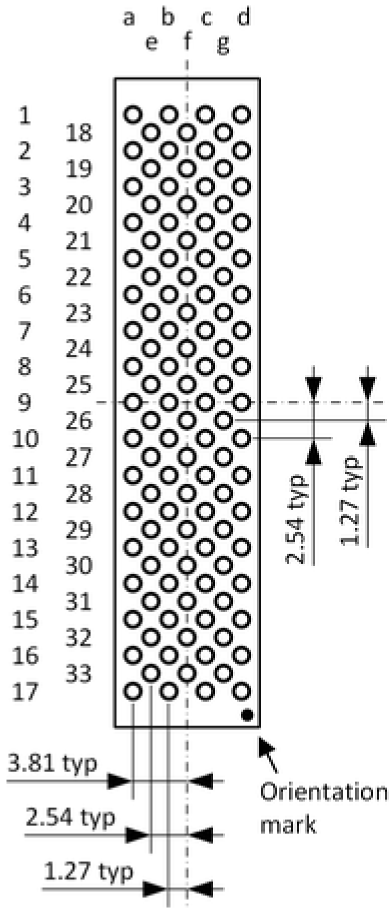
Very-high-density (VHD) pogo cable assembly

32 channels per pogo block

	a	e	b	f	c	g	d
1	GSA2		PA1		GND		PSA1
18		PB1		PSB1		GSB1	
2	PSA2		GND		PA2		GSA1
19		GSB2		PB2		PSB2	
3	GSA4		PA3		GND		PSA3
20		PB3		PSB3		GSB3	
4	PSA4		GND		PA4		GSA3
21		GSB4		PB4		PSB4	
5	GSA6		PA5		GND		PSA5
22		PB5		PSB5		GSB5	
6	PSA6		GND		PA6		GSA5
23		GSB6		PB6		PSB6	
7	GSA8		PA7		GND		PSA7
24		PB7		PSB7		GSB7	
8	PSA8		GND		PA8		GSA7
25		GSB8		PB8		PSB8	
9	GSA10		PA9		GND		PSA9
26		PB9		PSB9		GSB9	
10	PSA10		GND		PA10		GSA9
27		GSB10		PB10		PSB10	
11	GSA12		PA11		GND		PSA11
28		PB11		PSB11		GSB11	
12	PSA12		GND		PA12		GSA11
29		GSB12		PB12		PSB12	
13	GSA14		PA13		GND		PSA13
30		PB13		PSB13		GSB13	
14	PSA14		GND		PA14		GSA13
31		GSB14		PB14		PSB14	
15	GSA16		PA15		GND		PSA15
32		PB15		PSB15		GSB15	
16	PSA16		GND		PA16		GSA15
33		GSB16		PB16		PSB16	
17	GND		NC		GND		NC

Orientation mark •

Layout and dimension



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EK751HDB - High Density Pogo Cable for EK751 (8 x 16 channels)

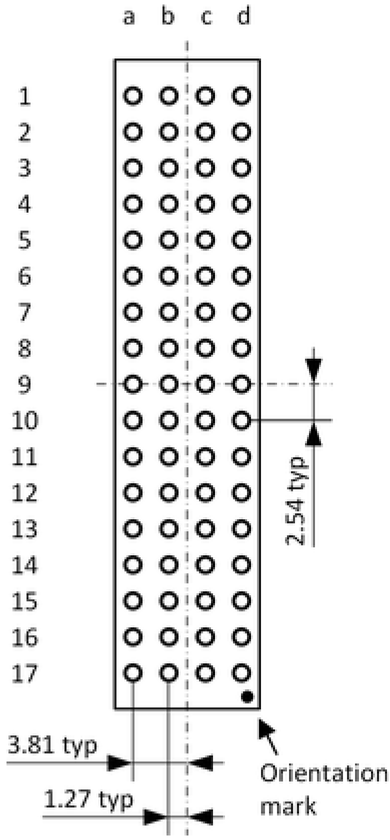
High-density (HD) pogo block

16 channels per pogo block

	a	**b**	**c**	**d**
1	GS2	P1	GND	PS1
2	PS2	GND	P2	GS1
3	GS4	P3	GND	PS3
4	PS4	GND	P4	GS3
5	GS6	P5	GND	PS5
6	PS6	GND	P6	GS5
7	GS8	P7	GND	PS7
8	PS8	GND	P8	GS7
9	GS10	P9	GND	PS9
10	PS10	GND	P10	GS9
11	GS12	P11	GND	PS11
12	PS12	GND	P12	GS11
13	GS14	P13	GND	PS13
14	PS14	GND	P14	GS13
15	GS16	P15	GND	PS15
16	PS16	GND	P16	GS15
17	GND	NC	GND	NC

Orientation mark •

Layout and dimension



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EK751VHDB - Very High Density Pogo Cable for EK751 (4 x 32 channels)

Very-high-density (VHD) pogo cable assembly

32 channels per pogo block

	a	e	b	f	c	g	d
1	GSA2		PA1		GND		PSA1
18		PB1		PSB1		GSB1	
2	PSA2		GND		PA2		GSA1
19		GSB2		PB2		PSB2	
3	GSA4		PA3		GND		PSA3
20		PB3		PSB3		GSB3	
4	PSA4		GND		PA4		GSA3
21		GSB4		PB4		PSB4	
5	GSA6		PA5		GND		PSA5
22		PB5		PSB5		GSB5	
6	PSA6		GND		PA6		GSA5
23		GSB6		PB6		PSB6	
7	GSA8		PA7		GND		PSA7
24		PB7		PSB7		GSB7	
8	PSA8		GND		PA8		GSA7
25		GSB8		PB8		PSB8	
9	GSA10		PA9		GND		PSA9
26		PB9		PSB9		GSB9	
10	PSA10		GND		PA10		GSA9
27		GSB10		PB10		PSB10	
11	GSA12		PA11		GND		PSA11
28		PB11		PSB11		GSB11	
12	PSA12		GND		PA12		GSA11
29		GSB12		PB12		PSB12	
13	GSA14		PA13		GND		PSA13
30		PB13		PSB13		GSB13	
14	PSA14		GND		PA14		GSA13
31		GSB14		PB14		PSB14	
15	GSA16		PA15		GND		PSA15
32		PB15		PSB15		GSB15	
16	PSA16		GND		PA16		GSA15
33		GSB16		PB16		PSB16	
17	GND		NC		GND		NC

Orientation mark •

Layout and dimension